



TxOne bootcamp

TXOne Global Training Team
Chris Van Den Abbeele

Wifi:
Password:

A very quick Who is who

- Name
- Experience with TxOne products ?

TxOne training matrix

Online / Self-paced

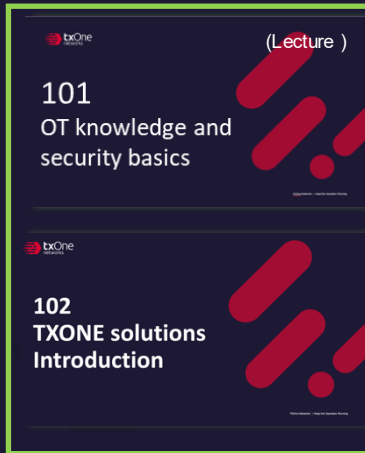
Hands-on, On-site

Certification: **Expert**
(by product line)
(Prerequisite: Specialist)

(200, self-paced)

Certification: **Associate**

Certification: **Specialist**
(Prerequisite: Associate)

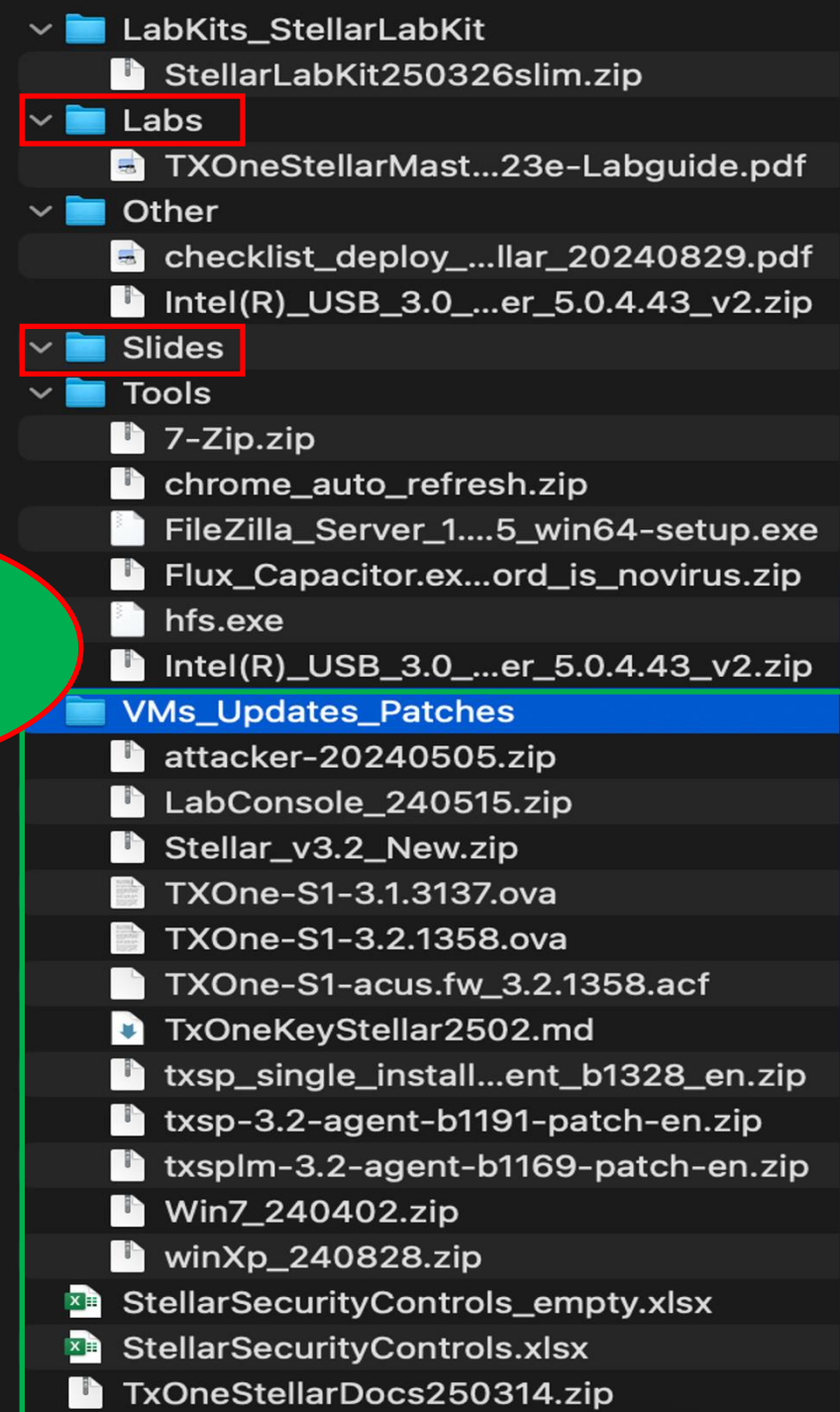
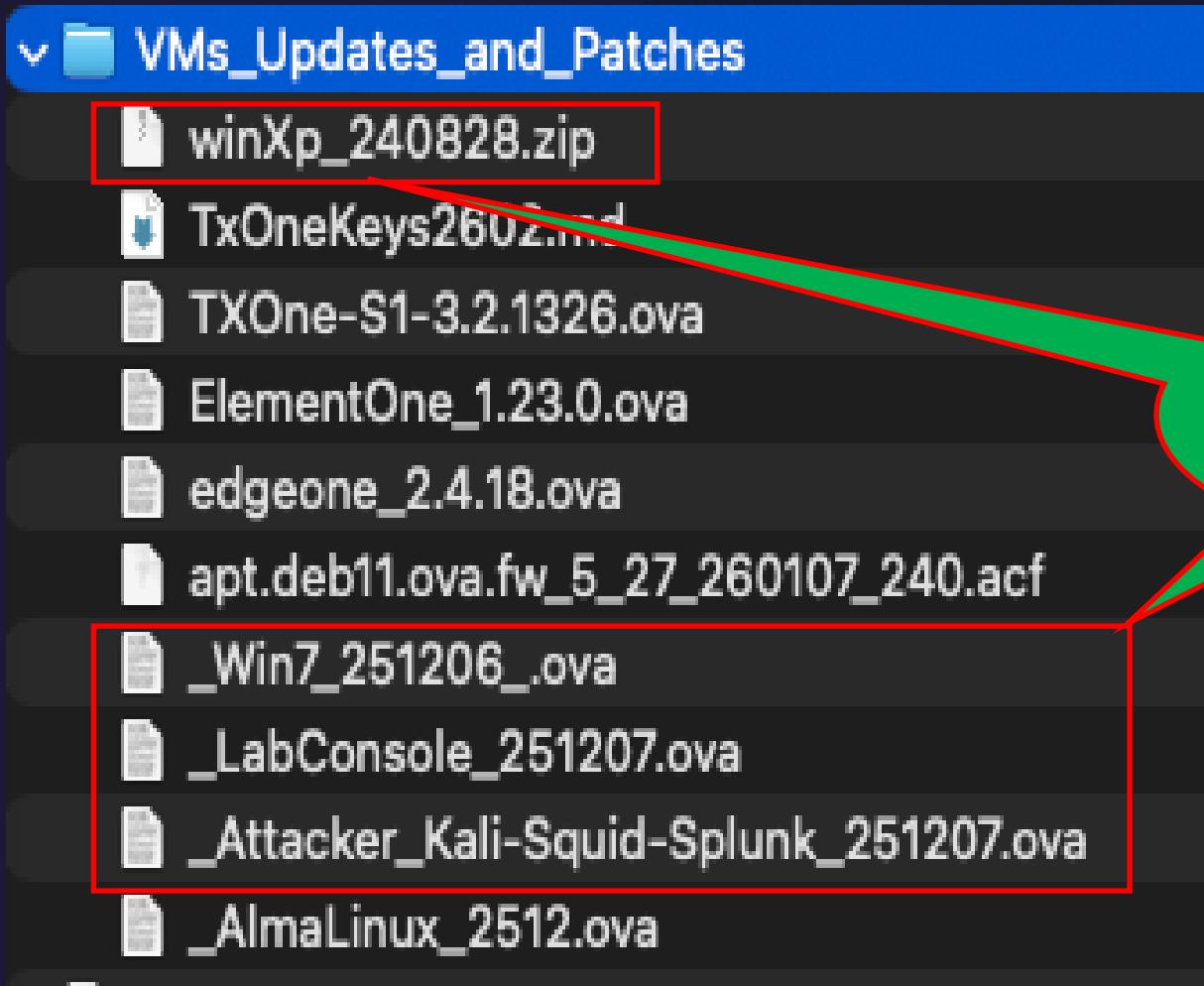


(400)

(300)

(350, hands-on,
this bootcamp)

Copy the entire USB stick to your laptop





350 TxOne Edge bootcamp

TXOne Global Training Team
Chris Van Den Abbeele

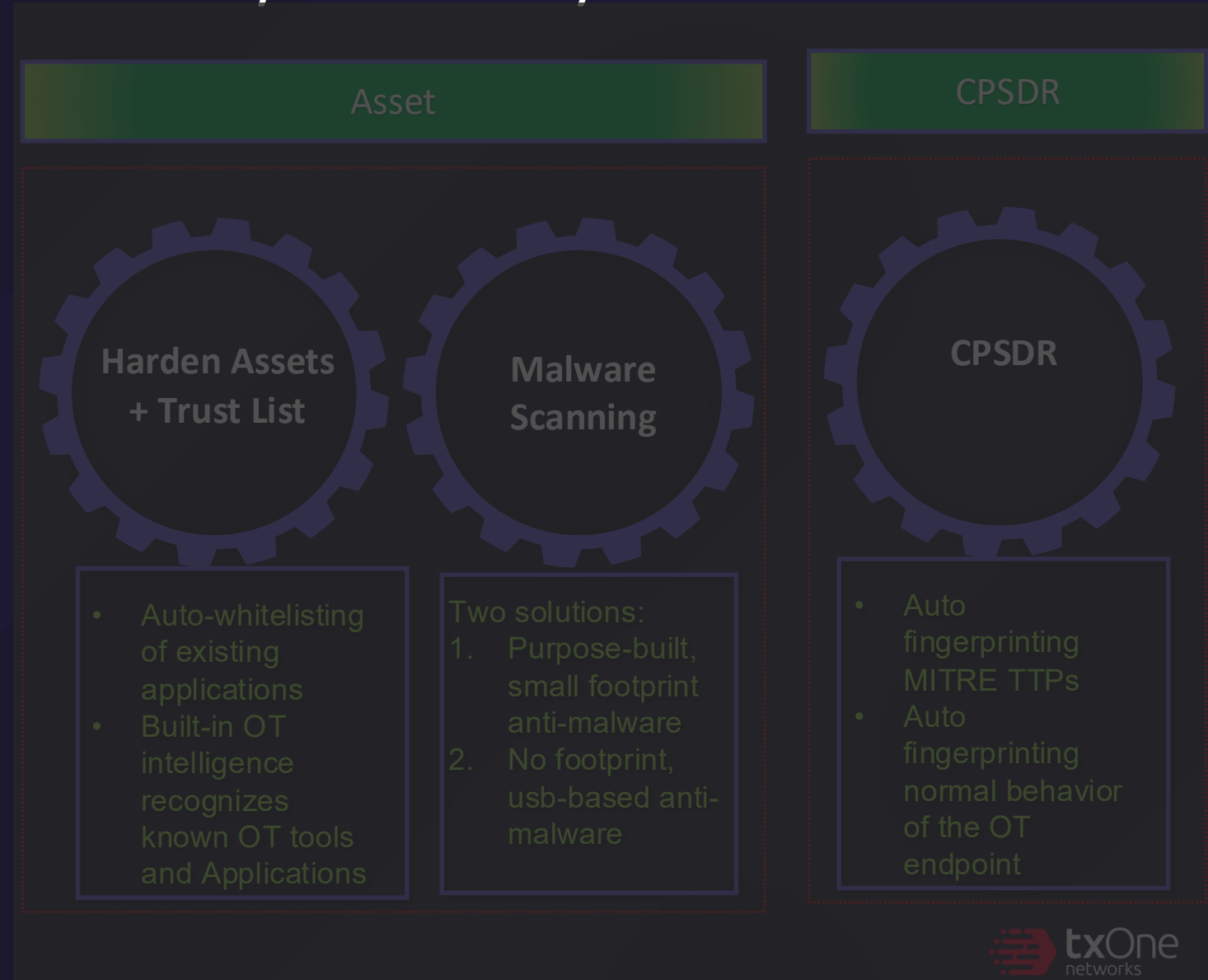
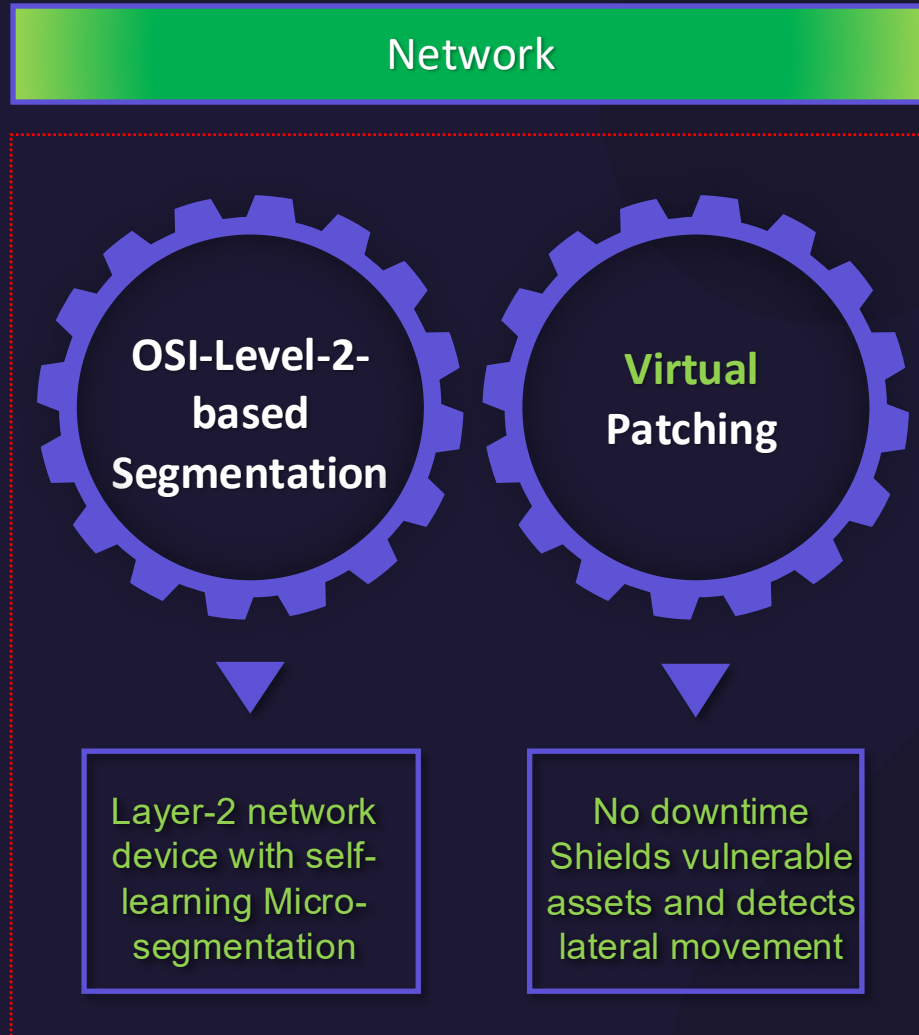
Agenda

- Industry Best Practices, OT Challenges, TxOne Zero Trust for OT (201 recap)
- Edge Product Suite (201 recap)
- Reference Deployment (202 recap)
- **Hands-on labs: Edge Deployment** (300)
- Edge Key Features and Use-cases (201 recap)
- **Hands-on labs: Edge configuration and policies** (400)
- Integrations and Technology Partners
- Administration RBAC and SSO
- Troubleshooting Edge (302)

(The windows VMs for this training are provided without a MS license. You need to add your own Microsoft license)

Industry Best Practices, OT Challenges, TXOne Zero Trust for OT

TXOne Endpoint Solution for ICS Cybersecurity Resilience



TXOne's Automated Approach for Edge



1. **Learn** the Operation

- Set Edge device in "**Policy Rule Auto-Learning**" mode
- Edge inspects network traffic, including OT Protocol commands and parameters.

2. **Build** baseline / policy

- Let Edge build the policy e.g. for 2 days or 2 weeks

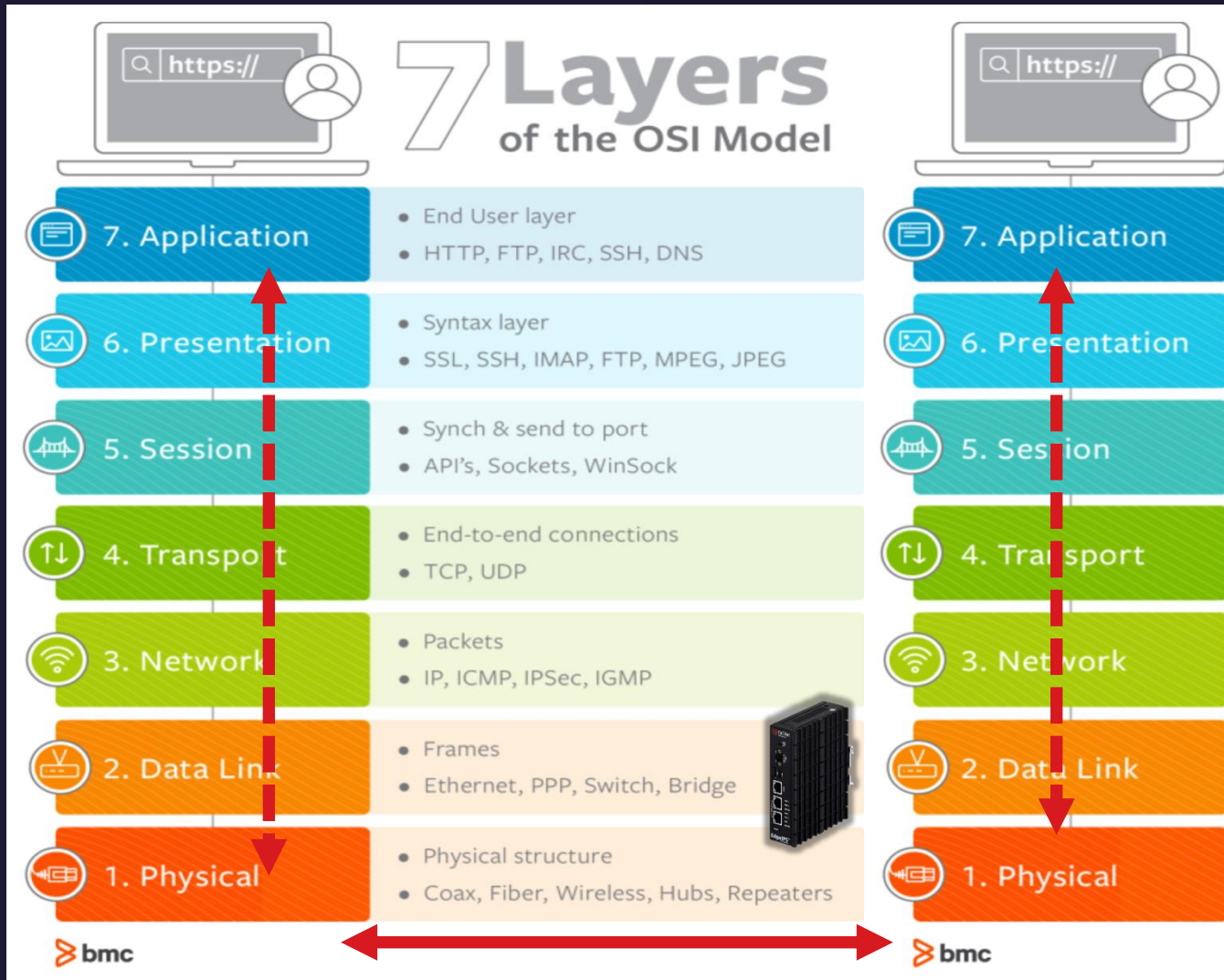
3. **Review** the policy

- Review baseline Policy and adapt if needed

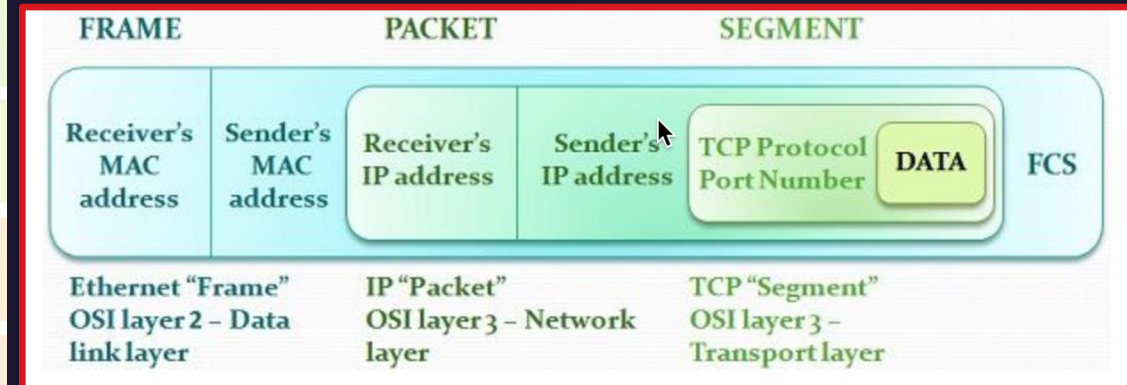
4. **Enforce** the policy

- Apply policy without impacting production
- Easy switching between "**monitoring**" and "**prevention**" mode

7 layers of the OSI model



EdgeIPS devices are Layer 2 networking devices that provide ***L2-L7 Visibility and Protection***





Edge Product Suite

Edge Product Family

EdgeOne

- Available management all Edge product in one console
- Network visibility
- Centralized policy management
- SIEM and XDR integration
- Security posture and report
- Multi-hypervisor support (Vmware /KVM/Hyper-V)



EdgeIPS LE-102

- 100Mbps+ performance
- 1 pairs Hardware bypass

EdgeIPS-103-Rugged

- 850Mbps+ performance
- 1 pairs hardware bypass
- OOB support

New Model

EdgeFire-1014-Rugged

- 850Mbps+ performance
- 2 WAN +10 LAN
- NAT / VPN

New Model

EdgeIPS-102-Rugged

- 200Mbps+ performance
- 1 pairs Hardware bypass

New Model

EdgeIPS-103F-Rugged

- 850Mbps+ performance
- 1 pairs Fiber hardware bypass
- OOB support

EdgeFire-1012-Rugged

- 200Mbps+ performance
- 2WAN +8 LAN
- NAT / VPN

EdgeIPS Pro-216 Rugged

- 1.8Gbps+ performance
- 8 pairs hardware bypass
- OOB support

EdgeIPS Pro-216 Commercial

- 1.8Gbps+ performance
- 8 pairs hardware bypass
- OOB support

New Model

EdgeFire-1014-Commercial

- 850Mbps+ performance
- 2WAN +10 LAN
- NAT / VPN

EdgeIPS Pro-212F

- 1.8Gbps Performance
- 4 Pairs Fiber with bypass
- 2 Pairs Copper with bypass
- OOB support

New Model

EdgeIPS Pro-2008

- 20Gbps Performance
- 4 Pair 10G copper with bypas
- OOB support

EdgeIPS Pro-2096

- 20Gbps Performance
- Up to 48 pairs hardware bypass
- OOB support

EdgeIPS Pro-1048

- 10Gbps Performance
- Up to 24 pairs hardware bypass
- OOB support

EdgeIPS Pro-732

- 7.2Gbps Performance
- 16 Pairs copper with bypass
- OOB Support

New Model

EdgeIPS Pro-2016F/4016F

- 20G/40Gbps performance
- 8 pairs 10G fiber with bypass
- OOB Support

New Model

All Data Sheets

<https://www.txone.com/data-sheets/>

Filter by tag 



Data Sheets

Jan 15, 2025

EdgeIPS Standard Series datasheet



Data Sheets

Jan 15, 2025

EdgeIPS Pro Series datasheet



Data Sheets

Jan 15, 2025

EdgeFire datasheet



Data Sheets

Jan 15, 2025



Data Sheets

Jan 15, 2025



Data Sheets

Dec 17, 2024

Edge Product Suite

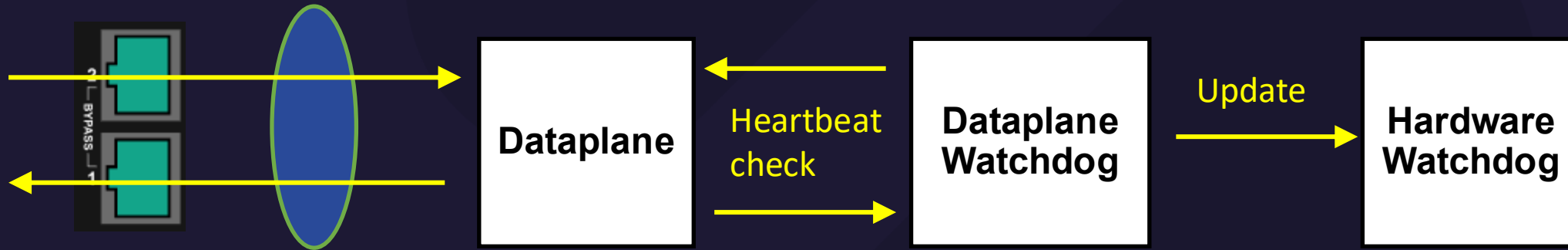


Features	EdgeIPS 103	EdgeIPS Pro 216	EdgeIPS Pro 1048/2096	EdgeIPS Pro 2008/2016F	EdgeFire	EdgeIPS LE
Throughput	850Mbps+	2Gbps+	10Gbps/20Gbps	20Gbps	200Mbps	
Network Segmentation	1	8	12/24/48/96			
IP/ Protocol White-listing	✓	✓				
Virtual Patching						
1. All devices do IDS/IPS 2. Devices with "PRO" in the name also do (network-)virus scanning 3. EdgeFire is a Layer3 device; all others are Layer2 devices 4. All devices, except the EdgeIPS-LE, can manage and control OT protocols 5. The mentioned throughput is a guaranteed minimum with all features enabled and for all ports combined						
Options		✓ Ruggedized or Commercial format	✓ Module-Based I/O Card	✓ 10Gbps interface / Fiber bypass (2016 F)	✓ NAT Firewall / Network Segmentation	✓

Hardware Bypass

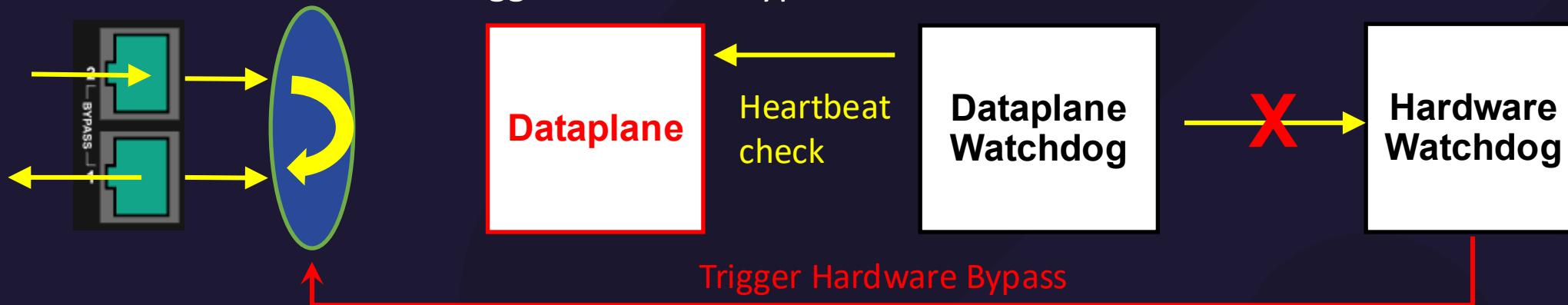
Normal Operation

“Dataplane Watchdog” monitors “Dataplane” process every second using heartbeat, and updates to “Hardware Watchdog”



Dataplane hangs / abnormal behavior

If Dataplane crashed, Dataplane Watchdog got no heartbeat, it stops updating to “Hardware Watchdog”. If problem persists for 6 seconds, Hardware Watchdog will trigger hardware bypass to be enabled





EdgeOne

EdgeOne – Virtual Machine for : Central Management and Extensibility



- Centralized Deployment for policies and patterns
- Supports up to **1,000** Network Segments
- Full Visibility of OT assets, operations and security events
- Instant Notification for critical events
- 3rd-Party Integration (SIEM, ICS Detection, TrendMicro Vision 1)
- Virtual Machine Platform Support
 - VMWare ESXi / Workstation, KVM, Hyper-V

Asset View

The screenshot displays the EdgeOne network management interface, specifically the Asset View section. The top navigation bar includes 'EdgeOne', 'Visibility', 'Nodes', 'Logs & Reports', 'Applications', and 'Admin'. The breadcrumb trail shows 'Visibility > OT Visibility > Asset View'. A '+ Add Filters' button is located on the left. The main area is divided into a left sidebar, a central asset grid, and a right-hand 'Information' panel.

Left Sidebar: A tree view of asset groups. 'EdgeIPS Pro' is selected, showing a sub-group 'EdgeIPS Pro 1048-G2...' with 100 assets. Other groups include 'Group Name 1' (2/2), 'Group Name 2' (3/3), 'Group Name 3' (1/3), 'Group Name 4' (4/5), 'Group Name 5' (5/5), 'EdgeIPS' (5), 'EdgeIPS LE' (3), and 'EdgeFire' (3).

Central Asset Grid: Displays a grid of asset icons. A context menu is open over the first asset, 'PLC Example Nr0', showing options: 'Copy Asset Hostname', 'Copy IP Address', 'Copy MAC Address', 'Create IP Object Profile', 'Bookmark Asset', and 'Configure Port Security'. The grid also shows 'Total Number of Selected Assets: 1/100'.

Information Panel: Provides details for the selected asset:

- Status:** Active - Not Bookmarked
- Asset Hostname:** PLC Example Nr 0
- Alias Name:**
- Comment:**
- IP Address:** 192.168.244.89
- MAC Address:** d2:71:aa:a4:22:49
- VLAN ID:** 20
- Asset Type:** OT Handheld Device
- OT Layer:** L3
- Vendor Name:** Rockwell-0
- Model Name:** LOGIX63521
- Firmware:** 101
- OS:** Wii system

Asset View

EdgeOne

Visibility

Nodes

Logs & Reports

Applications

Admin

?

txOne

networks

Visibility > OT Visibility > OT Layer View

All Edge Types

All Edge Groups

All Edge Hostnames

All Interfaces

Total Count

377

Edge Device Count

5

Asset Count

372

L5

L4

L3

L2.5

L2

L1.5

L1

L0

OPC Server

19

OT Appliance ...

16

OT Handheld ...

13

Others / Misc.

97

Printer in OT

29

Thin Client

7

VoIP Applianc...

6

Wireless Acce...

16

Others / Misc.

2

IPS-LE-102

IPS-Pro-1048

IPS-Pro-1048-G3

IPS-Pro-216

Industrial PC / ...

24

Network Appli...

37

Others / Misc.

UPS / PDU / PSU

5

Others / Misc.

3

Industrial Con...

29

Industrial Pro...

61

IPS-Pro-1048

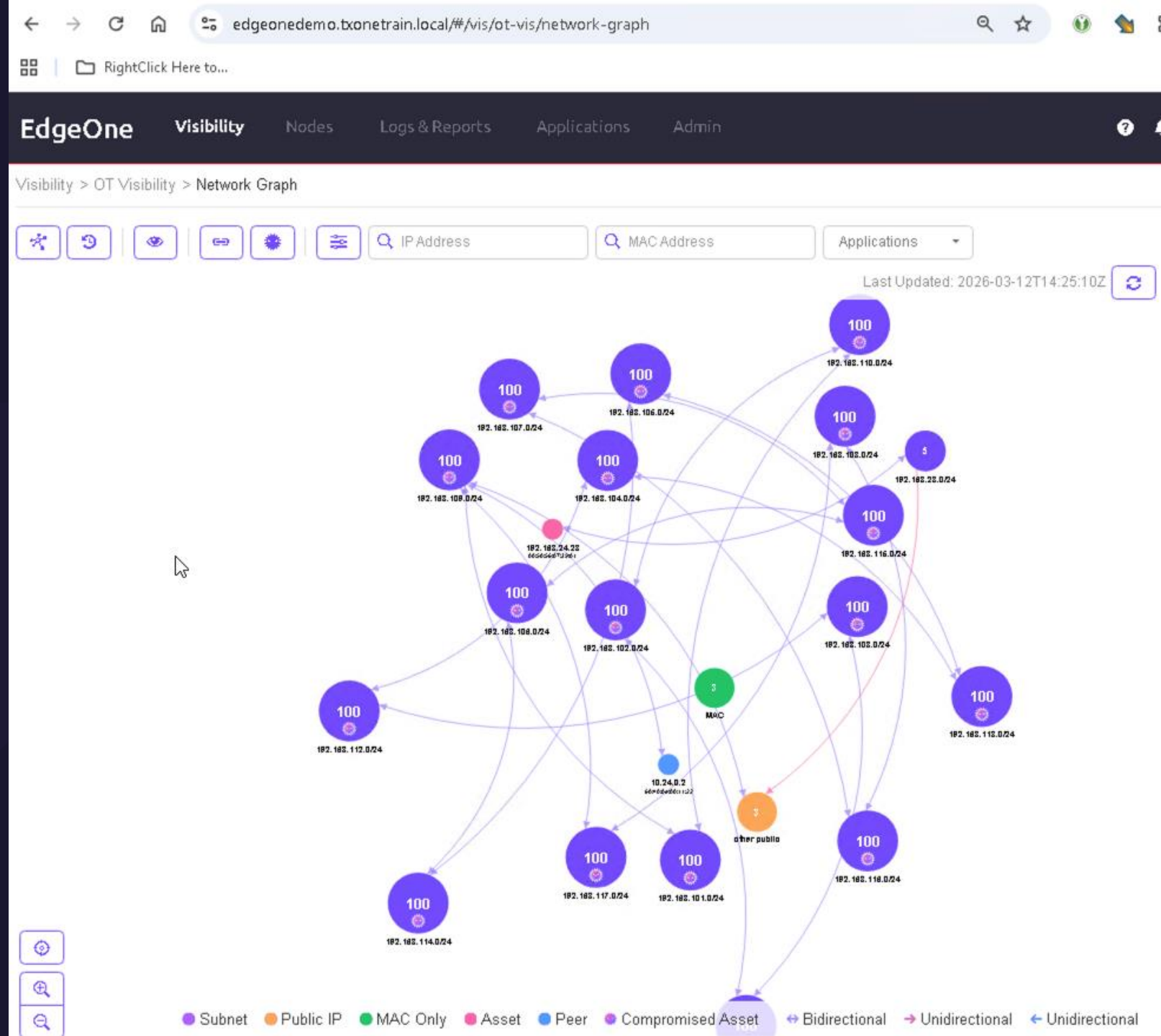
Industrial Driv...

4

Others / Misc.

3

Network Map View



Vulnerability View

MetLibre > OT Visibility > Vulnerability View

Download Vulnerability Detection Report

When managing CVEs, it's recommended to prioritize by severity so that limited resources are first allocated to vulnerabilities that could cause severe impact. Therefore, EdgeOne IPS patterns are designed to prioritize and provide best-eff...

Total 3337 Number of Vulnerabilities

Critical 10 Number of Vulnerabilities

High 130 Number of Vulnerabilities

Medium 314 Number of Vulnerabilities

Low 2883 Number of Vulnerabilities

Coverage Summary

- Covered 119
- Remediating 21

CVE-based View Asset-based View

Total Number of Items: 163 Search 1 / 2

All Asset Types All Severities All Risk Mitigation Statuses

Asset Hostname	IP Address	MAC Address	Asset Type	Legacy OS	OS Details	Severity
FAB124.037402	192.168.127.95	00:53:00:00:00:60	Others / Misc.	Legacy Windows (EOS)	Windows 7 / Server 2008, SP1, Build 7601	Critical: 5 → High: 73 → Medium: 155 → Low: 1
FAB124.060200	192.168.127.50	00:53:00:00:00:33	Industrial Controller - PLC/RTU	Legacy Windows (EOS)	Windows 10 / Server, Version 2004	Critical: 1 → High: 15 → Medium: 55 → Low: 67
FAB124.060817	192.168.127.19	00:53:00:00:00:14	Industrial PC / HMI / Panel PC	Legacy Windows (EOS)	Windows 7 / Server 2008, SP1, Build 7601	Critical: 5 → High: 73 → Medium: 155 → Low: 1
FAB124.085526	192.168.127.61	00:53:00:00:00:3e	Industrial Controller - PLC/RTU	Non-Legacy OS	Linux-based OS	Low: 2 →
FAB124.101483	192.168.127.21	00:53:00:00:00:16	Others / Misc.	Non-Legacy OS	Windows 10 Enterprise 2016 LTSC	Critical: 4 → High: 56 → Medium: 83 → Low: 15
FAB124.131206	192.168.127.72	00:53:00:00:00:49	Industrial Production Machines / Assets	Non-Legacy OS	Windows-based OS	Low: 1 →
FAB124.173693	192.168.127.44	00:53:00:00:00:2d	Network Appliance	Non-Legacy OS	Linux-based OS	Low: 1 →
FAB124.191545	192.168.127.31	00:53:00:00:00:20	OT Appliance (Desktop / Laptop)	Legacy Windows (EOS)	Windows 7 Professional SP1	Critical: 4 → High: 54 → Medium: 130 → Low: 6
FAB124.213667	192.168.127.27	00:53:00:00:00:1c	Others / Misc.	Legacy Windows (EOS)	Windows 7 / Server 2008, SP1, Build 7601	Critical: 5 → High: 73 → Medium: 155 → Low: 1
FAB124.223523	192.168.127.94	00:53:00:00:00:5f	Network Appliance	Non-Legacy OS	Linux-based OS	Low: 1 →
FAB124.229656	192.168.127.85	00:53:00:00:00:56	Industrial Production Machines / Assets	Legacy Windows (EOS)	Windows 7 / Server 2008, SP1, Build 7601	Critical: 5 → High: 73 → Medium: 155 → Low: 1
FAB124.293596	192.168.127.11	00:53:00:00:00:0c	OT Appliance (Desktop / Laptop)	Legacy Windows (EOS)	Windows 7 Professional SP1	Critical: 4 → High: 54 → Medium: 130 → Low: 6
FAB124.298620	192.168.127.99	00:53:00:00:00:64	Industrial Production Machines / Assets	Legacy Windows (EOS)	Windows 7 / Server 2008, SP1, Build 7601	Critical: 5 → High: 73 → Medium: 155 → Low: 1

Protection

15 *  Clear Filters

Total Number of Items: 30 1 / 1 < >  

Verities All Risk Mitigation Statuses

	Address	MAC Address	Asset Type	Legacy OS	OS Details	Severity	Risk Mitigation Status	Security Operation Mode	Policy Enforcement Operation Mo
FAB124.480204	192.168.127.75	00-53-00-00-00-4c	Industrial Production						
FAB124.513285	192.168.127.91	00-53-00-00-00-5c	Others / Misc.						
FAB124.553790	192.168.127.57	00-53-00-00-00-3a	Industrial Production						
FAB124.774248	192.168.127.77	00-53-00-00-00-4e	Wireless Access Po						
FAB124.844266	192.168.127.87	00-53-00-00-00-58	Wireless Access Po						
FAB124.954956	192.168.127.39	00-53-00-00-00-28	Others / Misc.						
FAB124.145445	192.168.127.60	00-53-00-00-00-3d	Wireless Access Po						
FAB124.227290	192.168.127.44	00-53-00-00-00-2d	Others / Misc.						
FAB124.432658	192.168.127.21	00-53-00-00-00-16	Wireless Access Po						
FAB124.567347	192.168.127.45	00-53-00-00-00-2e	Others / Misc.	Non-Legacy OS	Windows 10 / Server 2016, Version 1809	Low: 1 →	Partially Mitigated		
FAB124.725756	192.168.127.55	00-53-00-00-00-38	Others / Misc.	Non-Legacy OS	Windows 10 / Server 2016, Version 1809	Low: 1 →	Partially Mitigated		
FAB124.778565	192.168.127.32	00-53-00-00-00-21	Industrial Production Machines / Assets	Non-Legacy OS	Windows 10 / Server 2016, Version 1807	Low: 1 →	Partially Mitigated		
FAB124.814917	192.168.127.28	00-53-00-00-00-1d	Wireless Access Point and Controller	Non-Legacy OS	Windows 10 / Server 2016	Low: 1 →	Partially Mitigated		
FAB124.864109	192.168.127.97	00-53-00-00-00-62	Others / Misc.	Non-Legacy OS	Windows 10 / Server 2016	Low: 1 →	Partially Mitigated		
FAB124.871266	192.168.127.75	00-53-00-00-00-4c	Others / Misc.	Non-Legacy OS	Windows 10 Enterprise 2016 LTSS	Low: 1 →	Partially Mitigated		
FAB124.241574	192.168.127.78	00-53-00-00-00-4f	Wireless Access Point and Controller	Non-Legacy OS	Windows 10 / Server 2016	Low: 1 →	Partially Mitigated		
FAB124.449931	192.168.127.22	00-53-00-00-00-17	Others / Misc.	Non-Legacy OS	Windows 10 Enterprise 2016 LTSS	Low: 1 →	Partially Mitigated		
FAB124.458923	192.168.127.100	00-53-00-00-00-65	Others / Misc.	Non-Legacy OS	Windows 10 / Server 2016, Version 1809	Low: 1 →	Partially Mitigated		

Risk Mitigation Details (FAB124.145445, 192.168.127.60)

Mitigation Item	Status	Action
Edge group settings enabled	Yes	
[Denial of Service (DoS) Prevention] setting enabled	No	
[Policy Enforcement] setting enabled	No	
[CPSDR] setting enabled	Yes	

Close

Protection

Interface Settings

Interface

SLOT2:PORT2

Pair

PAIR1

Description

Settings

Pair Mode

Default

Security Operation Mode

Inline Mode

Offline Mode

Hardware Bypass Mode

Fail Open

Fail Close

Force Open

Click to [learn more](#) about how different policy rules are applied.

Policy Enforcement

Operation Mode

Monitor Mode

Prevention Mode

Device Rule Set

-

EdgeOne Rule Set

- Select -

DoS Prevention

Operation Mode

Monitor Mode

Prevention Mode

Edit DoS Settings

Edit

LFPT

Save

Cancel

27 TXOne Networks | Keep th

Protection

EdgeOne

Visibility

Nodes

Logs & Reports

Applications

Admin



Clear Filters

CVE-based View

Asset-based View

Clear Filters

Total Number of Items: 8

Search

All Asset Types

All Severities

Mitigated

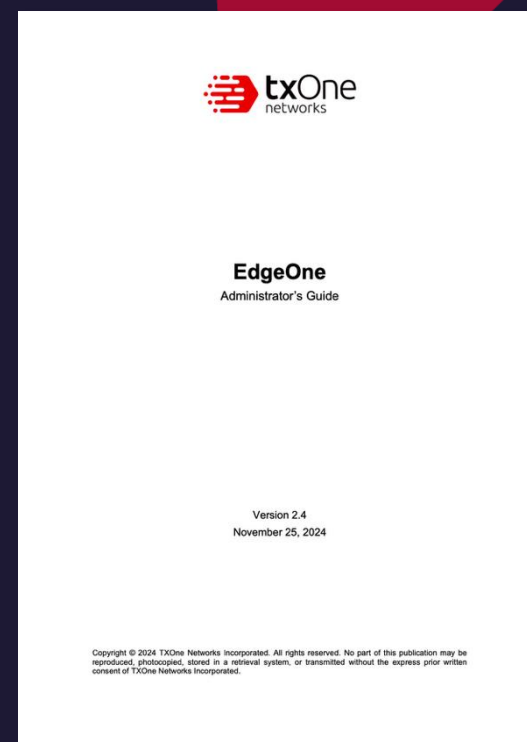
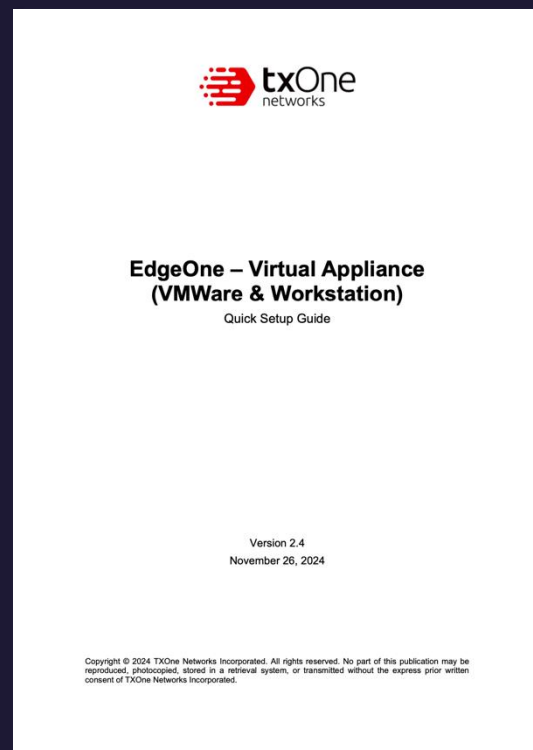


Asset Hostname	IP Address	MAC Address	Asset Type	Legacy OS	OS Details	Severity	Risk Mitigation Status	Security Operation
FAB124.101483 ⓘ	192.168.127.21	00:53:00:00:00:16	Others / Misc.	Non-Legacy OS	Windows 10 Enterprise 2016 LTSC	Low: 1 →	😊 Mitigated	🛡️ -
FAB124.351093 ⓘ	192.168.127.65	00:53:00:00:00:42	Others / Misc.	Non-Legacy OS	Windows 10 / Server 2016	Low: 1 →	😊 Mitigated	🛡️ -
FAB124.480204 ⓘ	192.168.127.75	00:53:00:00:00:4c	Industrial Production Machines / Assets	Non-Legacy OS	Windows 10 / Server 2016, Version 1607	Low: 1 →	😊 Mitigated	🛡️ -
FAB124.513285 ⓘ	192.168.127.91	00:53:00:00:00:5c	Others / Misc.	Non-Legacy OS	Windows 10 / Server 2016, Version 1809	Low: 1 →	😊 Mitigated	🛡️ -
FAB124.553790 ⓘ	192.168.127.57	00:53:00:00:00:3a	Industrial Production Machines / Assets	Non-Legacy OS	Windows 10 / Server 2016, Version 1607	Low: 1 →	😊 Mitigated	🛡️ -
FAB124.774248 ⓘ	192.168.127.77	00:53:00:00:00:4e	Wireless Access Point and Controller	Non-Legacy OS	Windows 10 / Server 2016	Low: 1 →	😊 Mitigated	🛡️ -
FAB124.844266 ⓘ	192.168.127.87	00:53:00:00:00:58	Wireless Access Point and Controller	Non-Legacy OS	Windows 10 / Server 2016	Low: 1 →	😊 Mitigated	🛡️ -
FAB124.954956 ⓘ	192.168.127.39	00:53:00:00:00:28	Others / Misc.	Non-Legacy OS	Windows 10 / Server 2016	Low: 1 →	😊 Mitigated	🛡️ -



Deployment

On USBstick:
TxOneDocs26xxxx.zip



EdgeOne server deployment

Download the right Virtual Machine image (for this training it is on the USB stick)

MyTXOne

Overview

Resources

Product Licensing

Software

Patterns

Documentation

Edge V1

ODC-VA

2023-07-18

ODC-VA Firmware(Full image upgrade)

1.5.24

Model: ODC-VA - full upgrade patch

File Size: 746MB

Download

Edge V1

EdgeIPS Std

2023-08-15

EdgeIPS Std Series Firmware for EdgeIPS-102(V1)

1.3.5

File Size: 66MB

Download

Edge V1

EdgeFire Std

2023-08-15

EdgeFire Std Series Firmware for EdgeFire-1012(V1)

1.2.11

File Size: 87MB

Download

Edge V2

EdgeOne

2023-07-07

TXOne EdgeOne - VMWare Image

2.1.14

Model: EdgeOne

File Size: 588MB

Download

Edge V2

EdgeOne

2023-07-10

TXOne EdgeOne - KVM Image

2.1.14

Model: EdgeOne

File Size: 612MB

Download

Edge V2

EdgeOne

2023-07-10

TXOne EdgeOne - KVM Image

2.1.14

Model: EdgeOne

File Size: 612MB

Download

MyTXOne Portal

Welcome to MyTXOne – the ultimate hub that revolutionizes the way you access, download, and manage files, documents, and licenses for your devices and services, consolidating everything you need all in one place, all at your fingertips.

Login > or Register >

Download

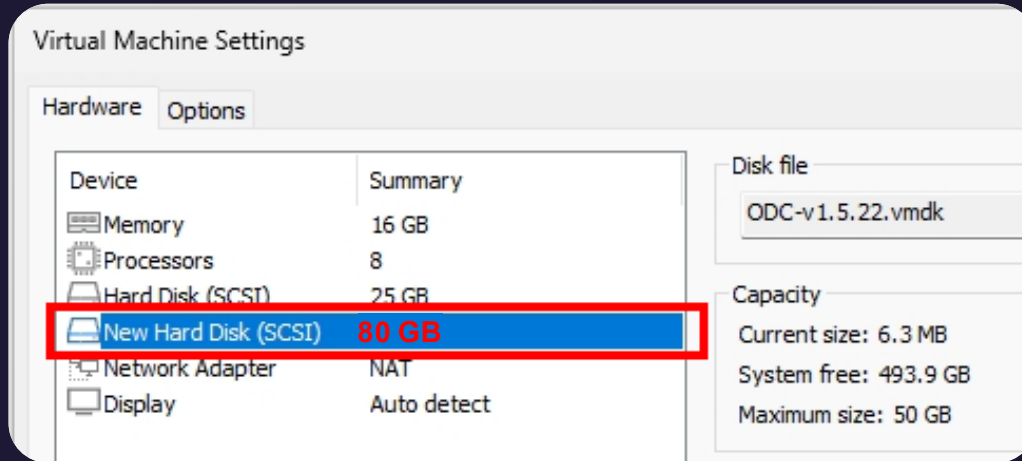
<https://my.txone.com>

Upgrade your account with a license key to get to the software section

More information at: <https://help.txone.com/docs/download-center-information>

EdgeOne server deployment (e.g. on Vmware)

- 1 Attach a 2nd Hard Disk to EdgeOne
80GB or larger



- 2 Boot the VM and login to the EdgeOne CLI

User: **admin** (in older versions, use root)

Password: **txone**

- 3 Boot the VM, login with admin/txone
Execute command "**oobe**" to change admin password to "12345678" -> P@ssw0rd

EdgeOne server deployment

3 Configure the IP address and DNS server

```
$ iface update eth0 --method static --address 192.168.23.40 --netmask 255.255.255.0 --gateway 192.168.23.2
```

```
$ resolv mode custom
```

```
$ resolv add 8.8.8.8
```

```
Debian GNU/Linux 11 ODC tty1
```

```
ODC login: root
```

```
Password:
```

```
Linux ODC 5.10.0-8-amd64 #1 SMP Debian 5.10.46-4 (2021-08-03) x86_64
```

```
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.
```

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.
```

```
Last login: Thu Feb 9 14:31:46 UTC 2023 on tty1
```

```
uShell, version v1.7.9
```

```
If you want to exit this shell, please type `exit` or `Ctrl-D`.
```

```
$ iface update eth0 --method static --address 192.168.23.20 --gateway 192.168.23.2 --netmask 255.255.255.0
```

```
Interface settings are changed. Please restart the interface and reboot this server to take effect.
```

```
$ resolv mode custom
```

```
$ resolv add 8.8.8.8
```

```
8.8.8.8 is added
```

```
$
```

EdgeOne server deployment

4 Reboot EdgeOne

(iface restart eth0)

(iface ls)

reboot

```
If you want to exit this shell, please type `exit` or `Ctrl-D`.
$ iface restart eth0
Successfully restart! Please reboot this server to take effect.
$ iface ls
[
  {
    "Name": "lo",
    "Family": "inet",
    "Method": "loopback"
  },
  {
    "Name": "eth0",
    "Family": "inet",
    "Method": "static",
    "Address": "192.168.23.20",
    "Netmask": "255.255.255.0",
    "Gateway": "192.168.23.2"
  }
]
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:ef:c1:77 brd ff:ff:ff:ff:ff:ff
    altname enp2s0
    altname ens32
    inet 192.168.23.20/24 brd 192.168.23.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:feef:c177/64 scope link
        valid_lft forever preferred_lft forever
```

5 Verify connectivity

dx au

```
If you want to exit this shell, please type `exit` or `Ctrl-D`.
$ dx au
start to test the following DNS servers:
8.8.8.8

OK!
start to connect the AU server...
OK!
$
```

EdgeOne server deployment

6 Check the status of the server

\$ **env ls**

```
If you want to exit this shell, please type `exit` or `Ctrl-D`.
```

```
$ env ls
```

```
Hostname: ODC
```

```
Status: RUNNING
```

```
Product Serial Number: 8ff54474-ddd6-11ed-b09a-000c29efc177
```

```
Version: 1.5.22
```

```
External IP: Not Set
```

```
DPI Engine Version: 2.0.14.d5d4c7
```

```
DPI Pattern Version: SDP_211215_12
```

```
Service Network: config: 169.254.0.0/16, network: [169.254.0.0/24, 169.254.1.0/24, 169.254.2.0/24]
```

```
Boot Time: 2023-04-21 02:13
```

```
$ _
```

Required *listening* ports of EdgeOne

EdgeOne Service	TCP	UDP	Description
Command Channel	7590		Command & logs channel (<i>persistent tunnel</i> from each Edge device)
Web Console	443		Access to the Edge Management Web Console
(SSH)	(22)		(Access Console via SSH)
Remote Access	9001		EdgeOne <i>proxies</i> the connection from the user's browser on port 9001 into the persistent tunnel that has been opened by the edgeIPS devices. This gives the user access to the webUI of the EdgeIPS even if those devices are behind NAT
NTP	123		registered EdgeIPS devices use EdgeOne as an NTP time-source
SNMP	161		from SNMP management to Edge/EdgeOne

Outgoing connections by EdgeOne (for Pattern Updates etc...)

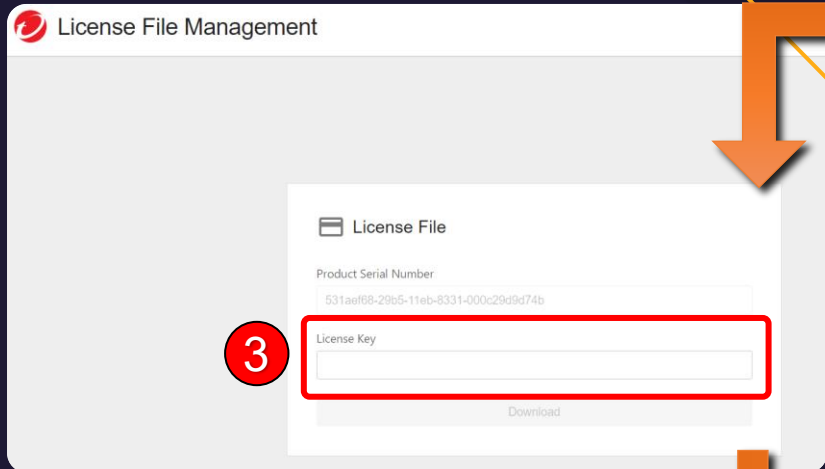
(Air-gapped networking is also supported)



Active Update Service	TCP	Description
odc.cs.txone-networks.com	443	TXOne Active Update Server
txone-component-prod.s3.amazonaws.com	443	Services for downloading signature and firmware

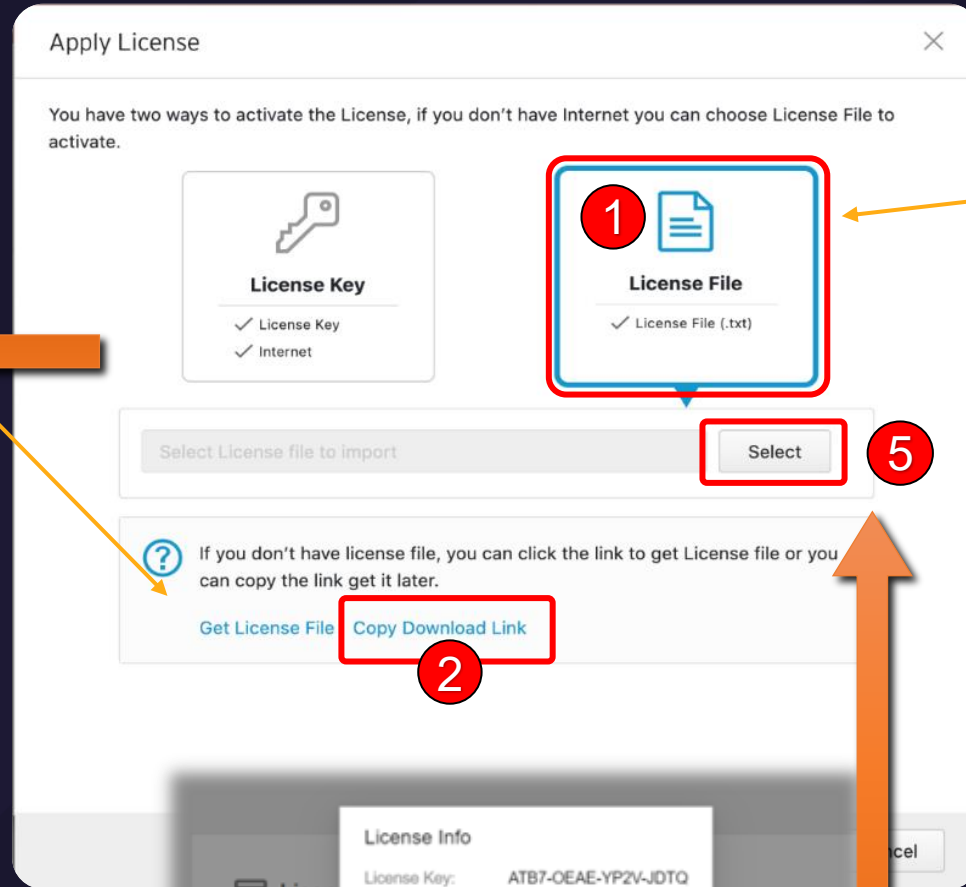
Offline license activation

Click on “**Copy Download Link**” and copy that link to a workstation that has Internet Connectivity



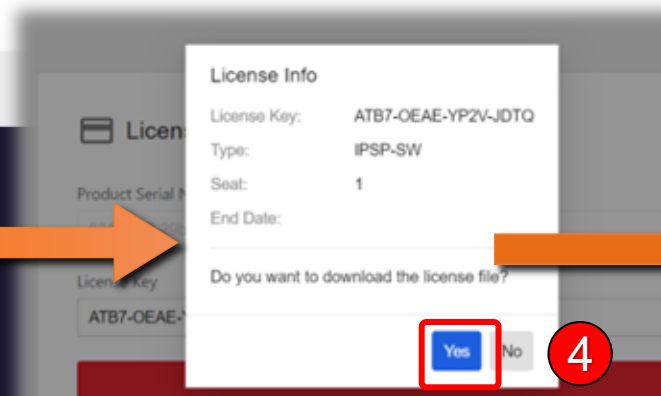
Open the Download link on a workstation that does have internet connectivity. This will open a web page.

The web page will ask for a License Code. Enter it



Choose the “**License File**” option

Upload the license file (from your workstation to the EdgeOne)



Download the license file And copy it to a workstation that can connect to the EdgeOne Web console

EdgeOne Licenses

- An EdgeOne Node License is required to activate the EdgeOne functions
- **Additional licenses** are required for each type of physical Edge device
- License types:

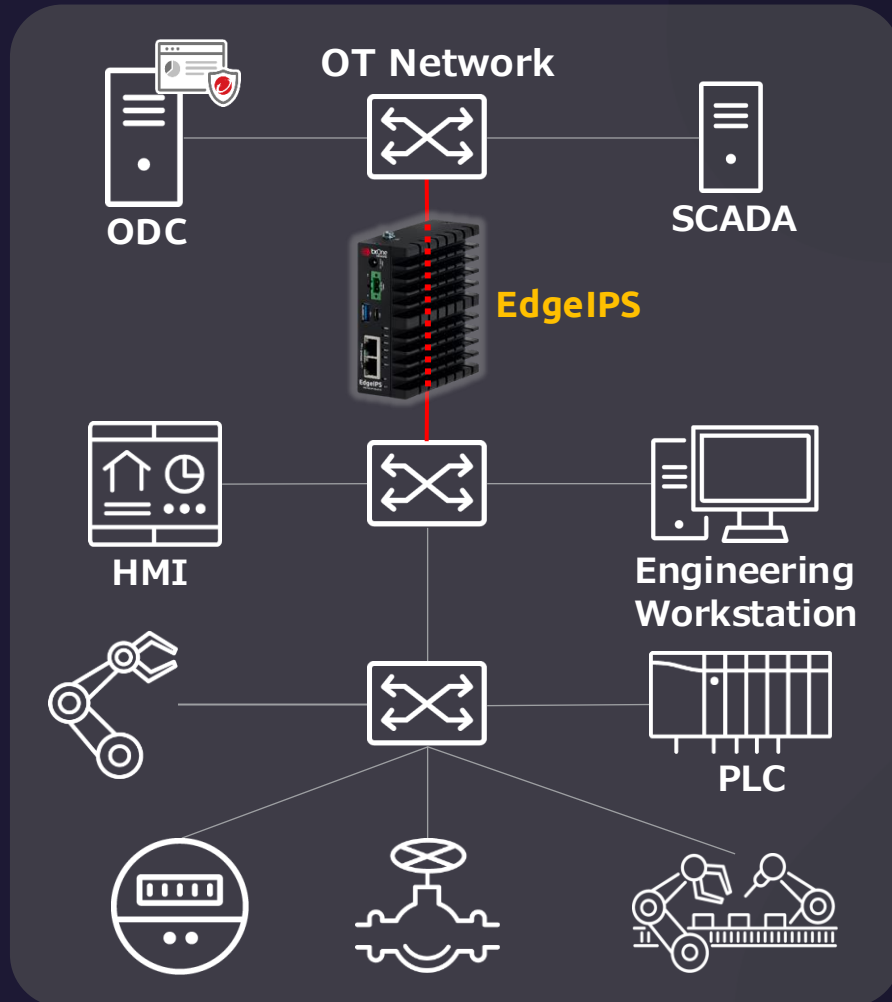
License Type	Scope	When expired
EdgeOne Node	- Enable node management functions - Enable EdgeOne active updates - Provide technical support	- Stop connections between EdgeOne and the managed devices. - User can view and query the existing logs.
EdgeIPS Software	- Enable Edge Series pattern / firmware update via EdgeOne - Provide technical support	- Stops pattern / firmware updates - Stops customer support. - Edge devices security functions remain available as is.
EdgeFire Software		
EdgeIPS Pro Software		

On the EdgeIPS Pro, each module counts for 12 seats

Where to put the Edge Devices in the network (recap)

EdgeIPS: Inline mode

Inline mode



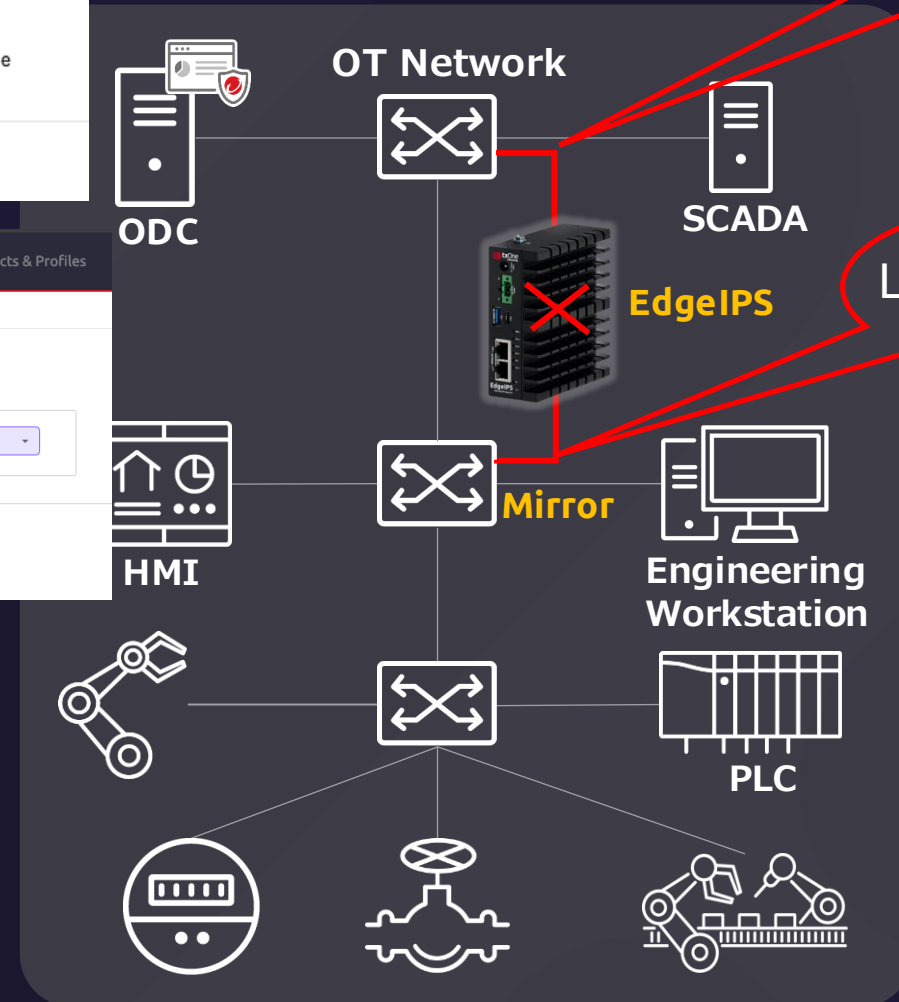
EdgeIPS

System	Visibility	Device
Security > Security General Settings		
Security Operation Mode		
☒ Inline Mode ☐ Offline Mode		
Save Cancel		

EdgeIPS	Visibility	Device	Objects & Profiles
Security > General Settings > Security Mode Settings			
Security Operation Mode			
☐ Inline Mode ☒ Offline Mode			
Management Port PORT1 MGMT ✓ PORT1 PORT2			
Save Cancel			

vs Offline mode

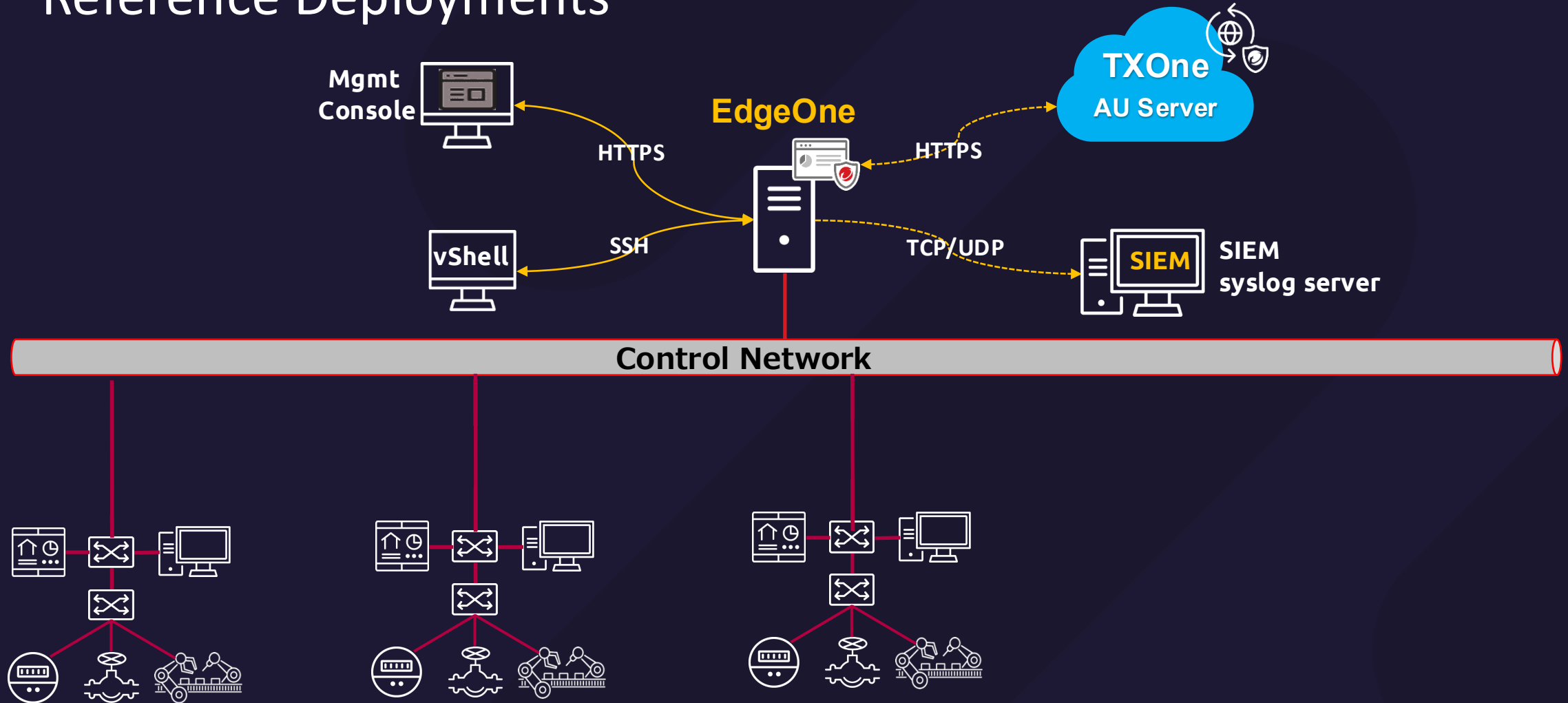
Offline mode



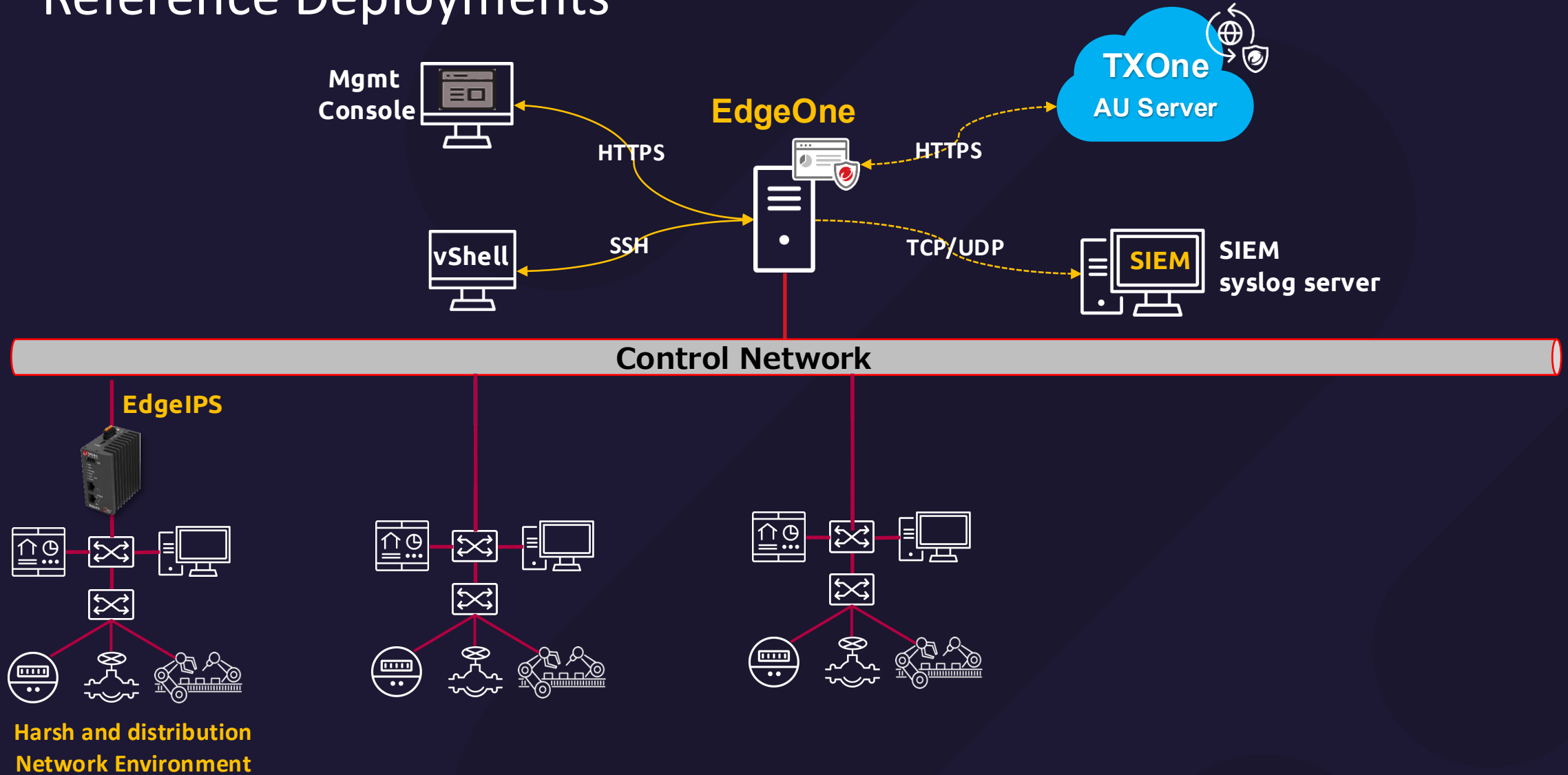
Management

Listen for traffic

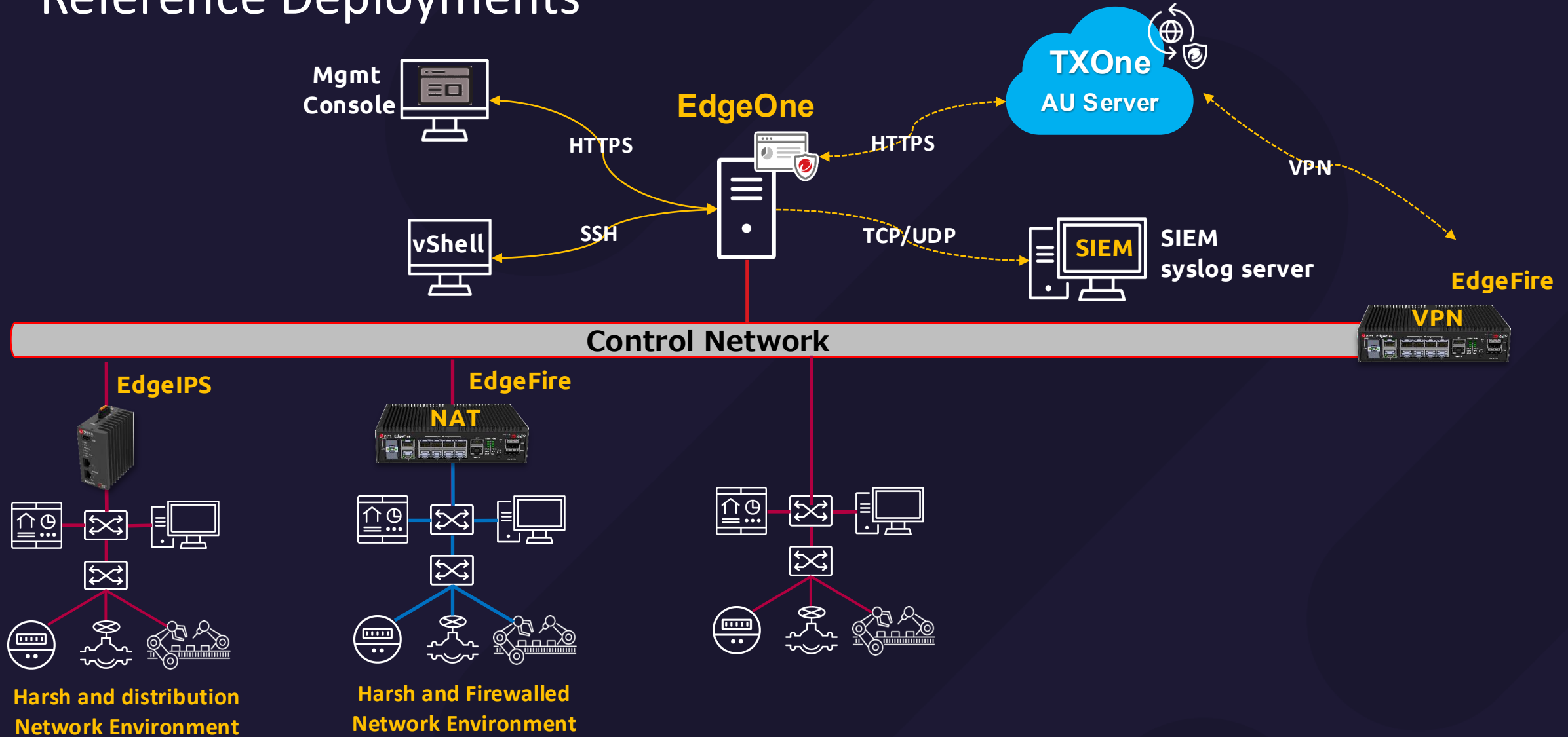
Reference Deployments



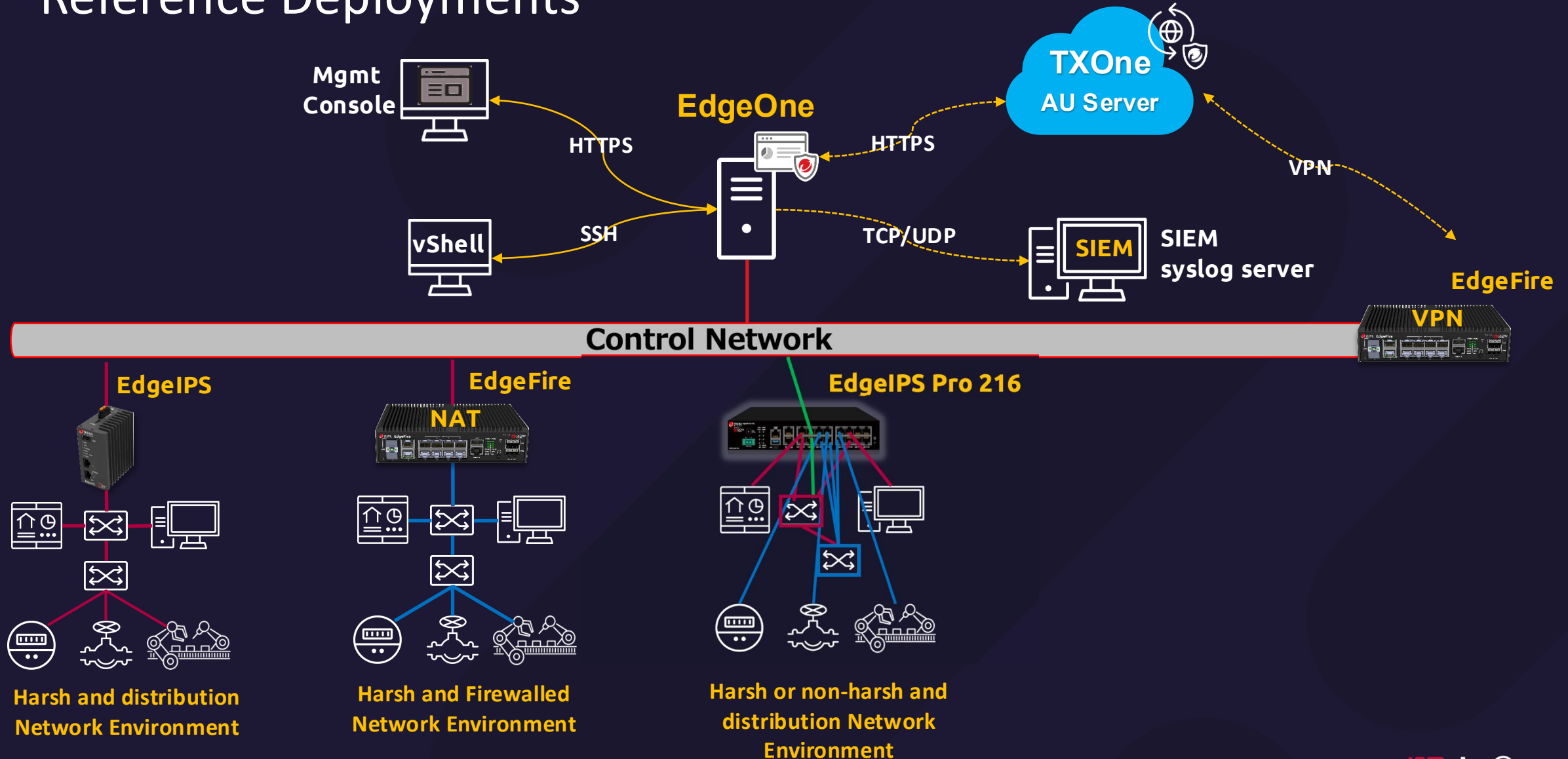
Reference Deployments



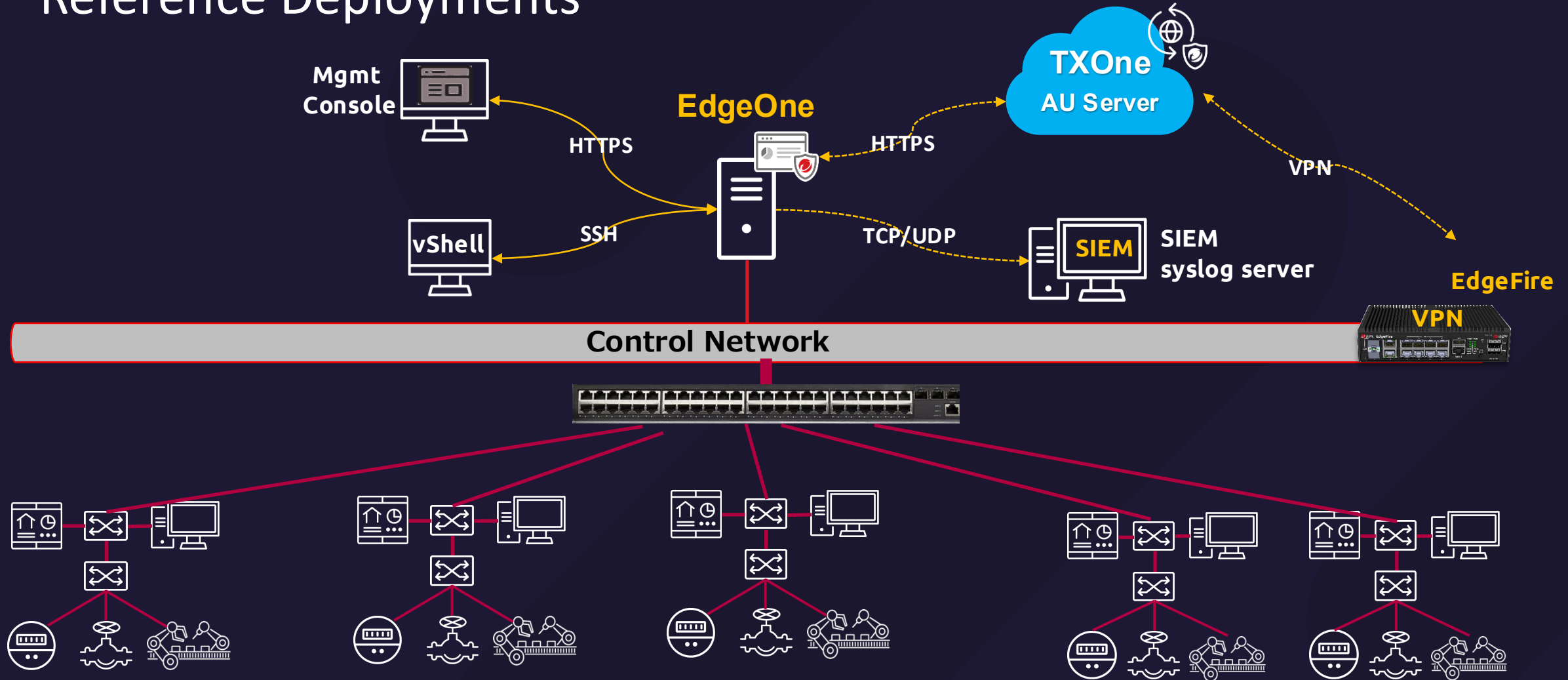
Reference Deployments



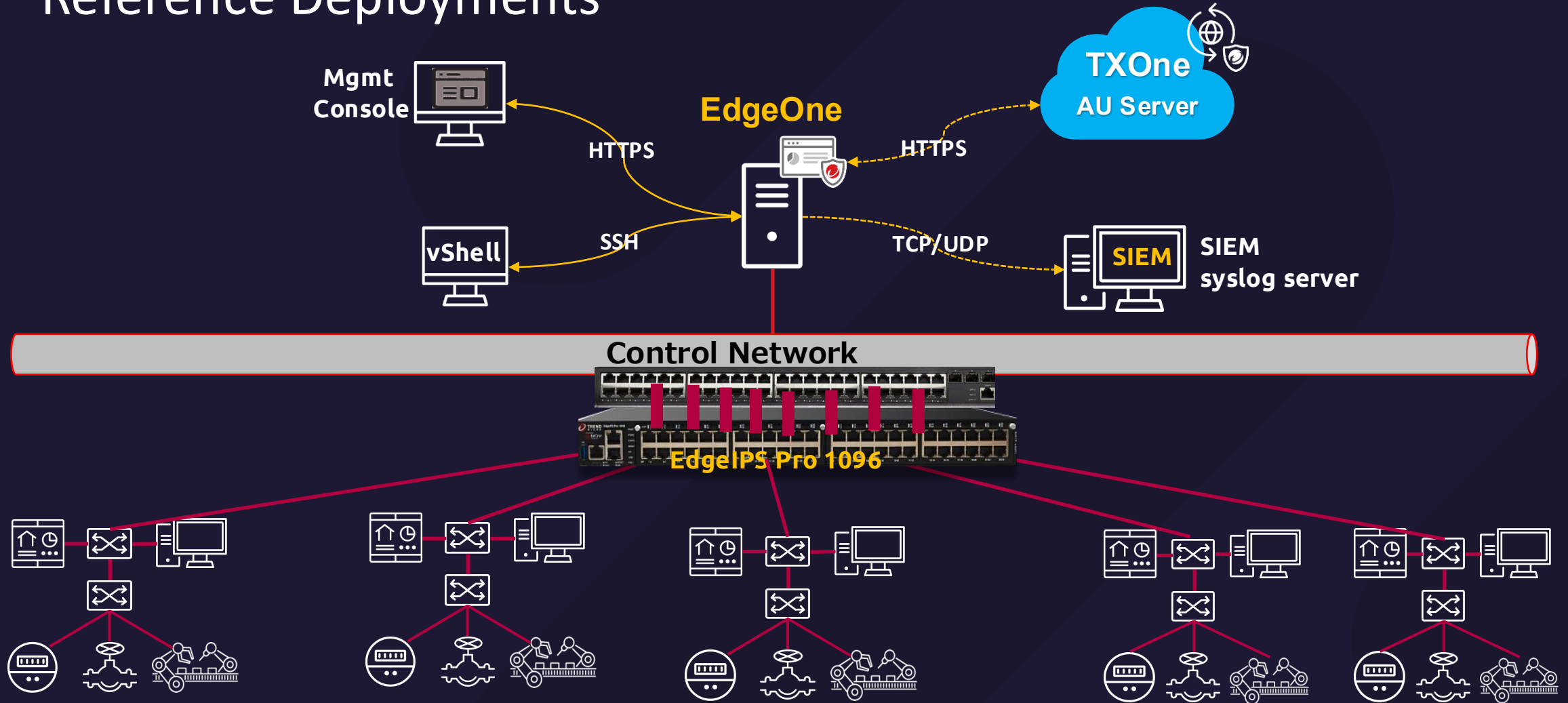
Reference Deployments



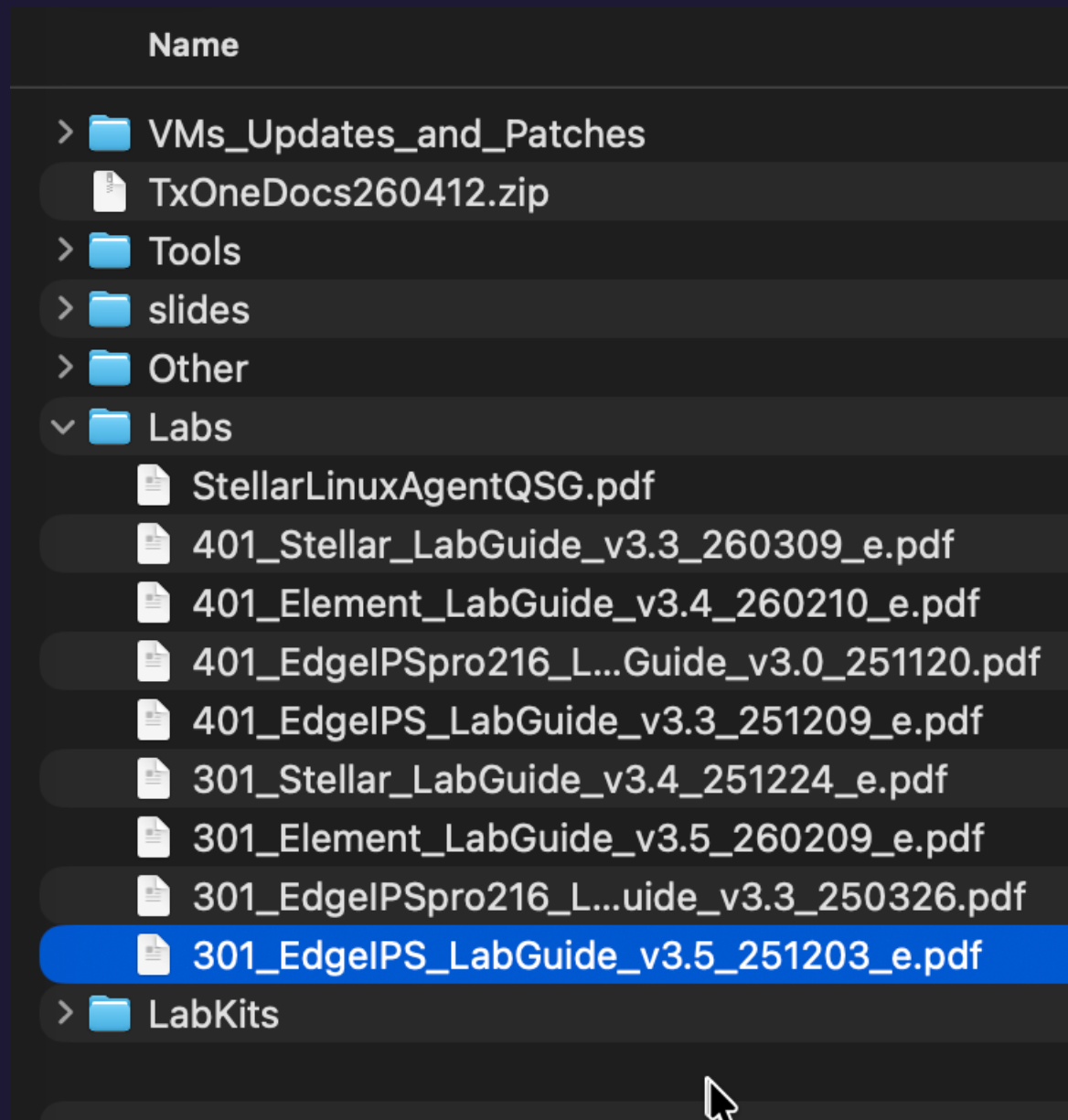
Reference Deployments



Reference Deployments



Deployment in the hands-on Labs



Virtual Machines Images

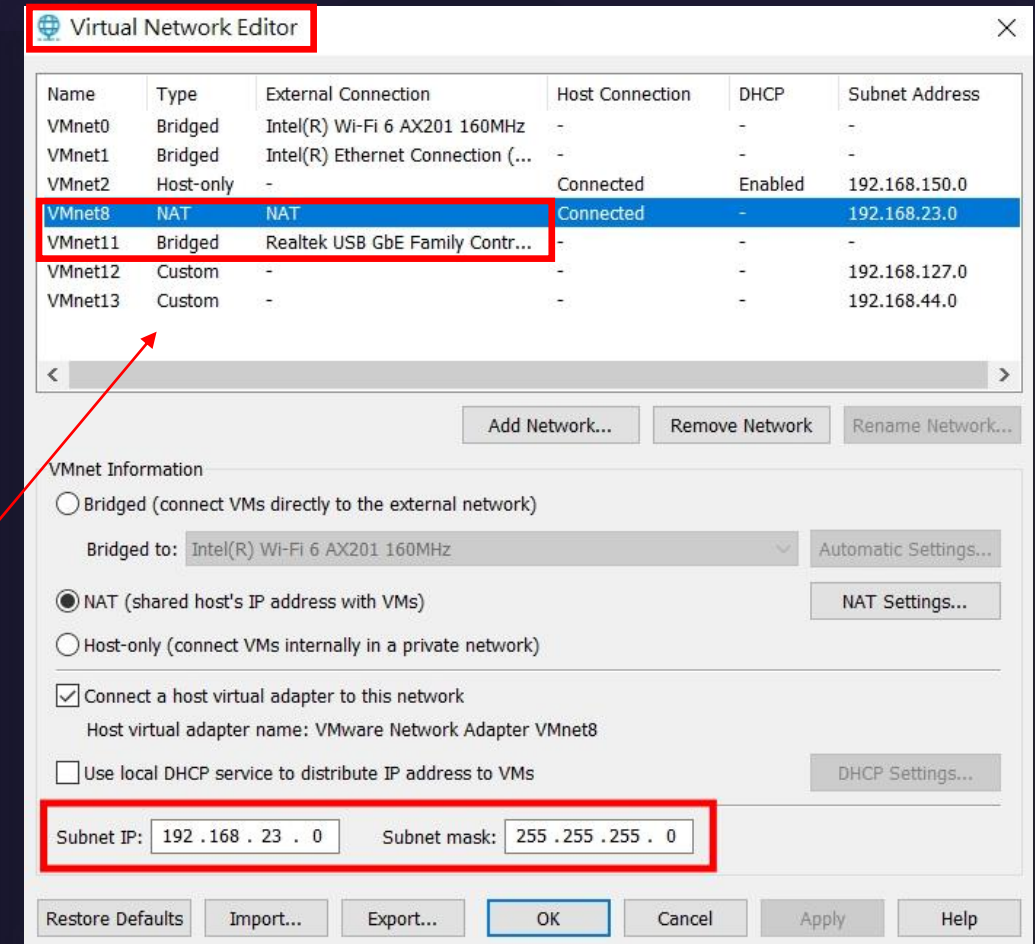
(Your laptop should have 16GB of RAM)

ZIP filename	VMware Image Name	Hardware Specification	Purpose
LabConsole_XXXXX.zip	LabConsole_XXXX	CPU: 2 Cores RAM: 4GB Hard Disk: 100G Network: Bridged to VMnet11	Web Console for EdgeOne, EdgeIPS, EdgeFire
EdgeOne	edgeone_2.x.y.z.ova	CPU: 8 Cores RAM: 16GB (can be reduced to 8GB) Hard Disk: 25G/80G Network: Bridged to VMnet11	EdgeIPS EdgeFire Management
Win7_XXXX.zip	Win7_XXXX	CPU: 2 Cores RAM: 2GB Hard Disk: 60G Network: Bridged to VMnet11	Target victim for EdgeIPS, EdgeFire
Kali (attacker)	Attacker_XXXX	CPU: 4 Cores RAM: 2GB Hard Disk: 80G Network: Bridge to VMnet11	

VMWare virtual networks for the labs

(See lab guides for more details)

- **NAT** network on **192.168.23.0 /24**
 - *First*, when the VMs need *internet connectivity*, we will connect them to this network (e.g. activating licenses, updating patterns,...)
- **Bridged** network on your UTP connector (typically this is **Vmnet11**)
 - *Then* we will change the connection from the VMs to the bridged network on your UTP port for communication with the EdgeIPS and the other Host
- **Important note:**
If you already have a network that is bridged to an "**Auto**" network adapter, use that one and *remap* it to your UTP adapter. **DO NOT USE "AUTO"**.



Lab 3

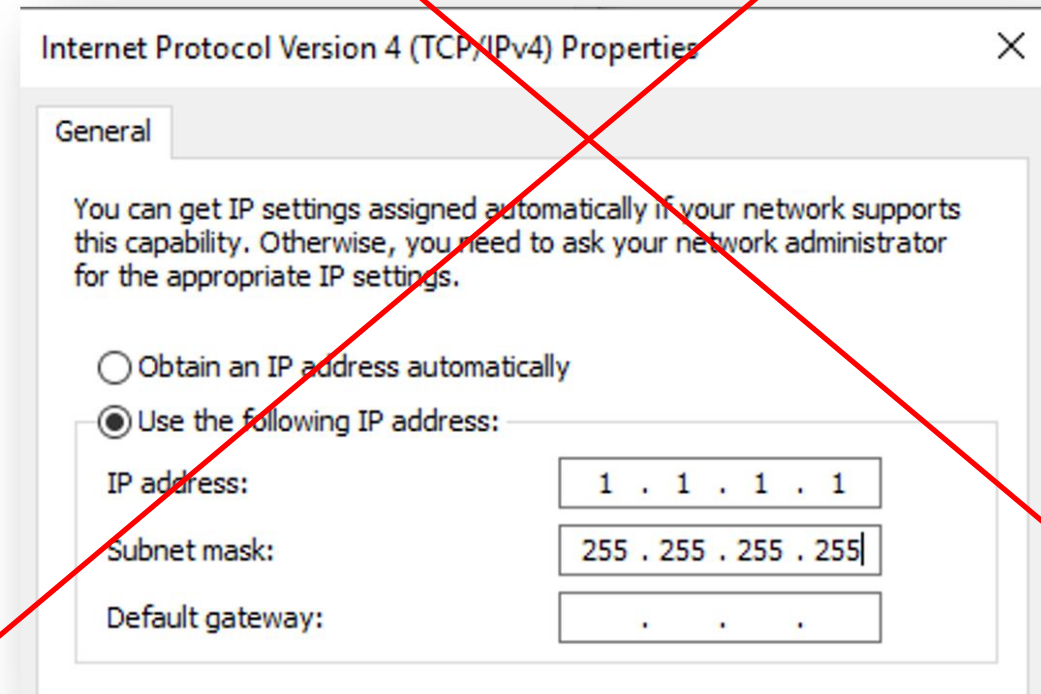
For better isolation between the VM network and the host network, the lab guide suggests you to assign a static, bogus, local IP to the network cards of your host

If you decide to do that, then do not forget to change it back to DHCP when you are back at work, or you will not have any network connectivity.

It may be easier to **just skip this step.**

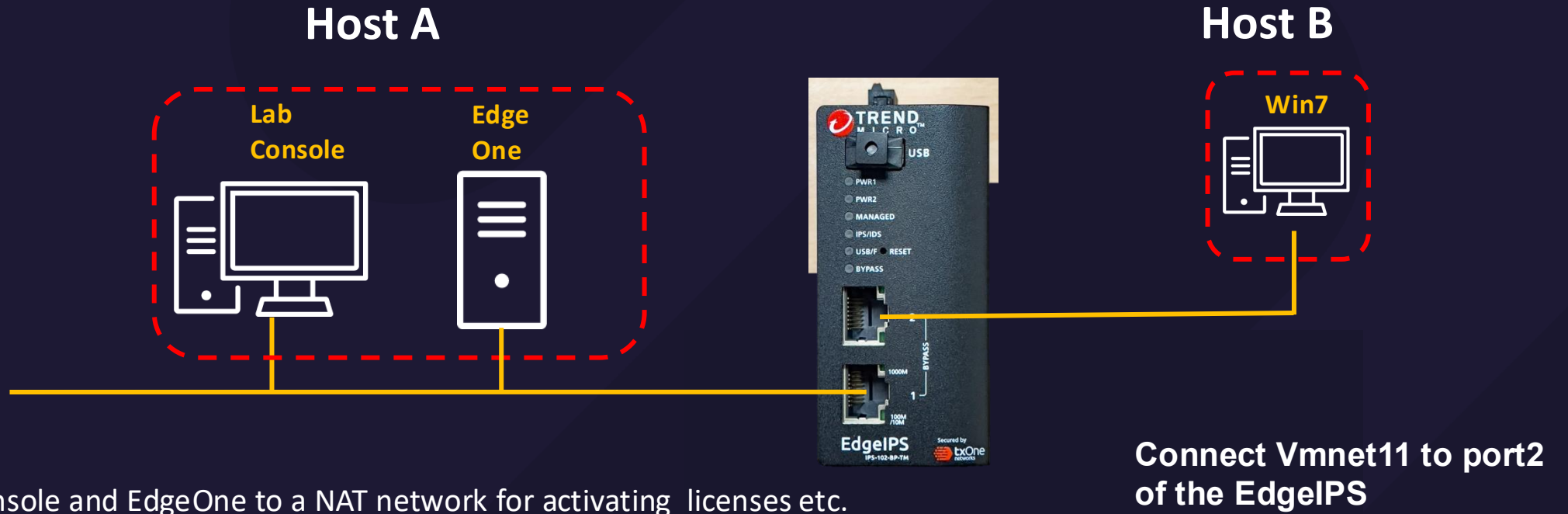
2. Assign physical host IP address 1.1.1.1/32 for Host A LAN Adapter and physical host IP address 1.1.1.2/32 for Host B LAN Adapter. This is for Lab network traffic isolation purpose.

Make sure network traffic will route thru EdgeIPS



Basic implementation training lab environment

(the first labs you will do individually, then you will team-up with a partner)



Connect LabConsole and EdgeOne to a NAT network for activating licenses etc.

Connect them to VMnet11 and then to port1 of EdgeIPS102

For EdgeIPS103: first connect to the Management port , when configuring the device, then switch to Port1 when sending data

Edge Implementation Labs

"301EdgeLabGuideTxOneSpecialist-...pdf"

("301 EdgeIPS pro216.....)

Lab1: EdgeOne Deployment.....

Lab2: Initial setup / upgrade of EdgeOne and web UI walk thru.....

Lab3: EdgeIPS Hardware, Installation and Lab setup.

Lab4: EdgeIPS device initialization

Lab5: Register EdgeIPS with EdgeOne for centralized management....

Lab6: Device Group Management.....

Lab7: Policy Enforcement Configuration

Lab8: EdgeIPS Policy Rule Auto-Learning

~~Lab9: EdgeFire initialize settings and registration with EdgeOne.....~~

~~Lab10: EdgeFire NAT Settings~~

The first labs, you do **individually**,
then you will **team-up** with a partner
(the labguide will tell you when)

The OVA VMs have no **snapshot**; just
make one

The licenses are in **txonekeys26xx.md**
in the VMs folder

Install

edgeone_2.4.18.ova

(in the VMs folder) and

upgrade to

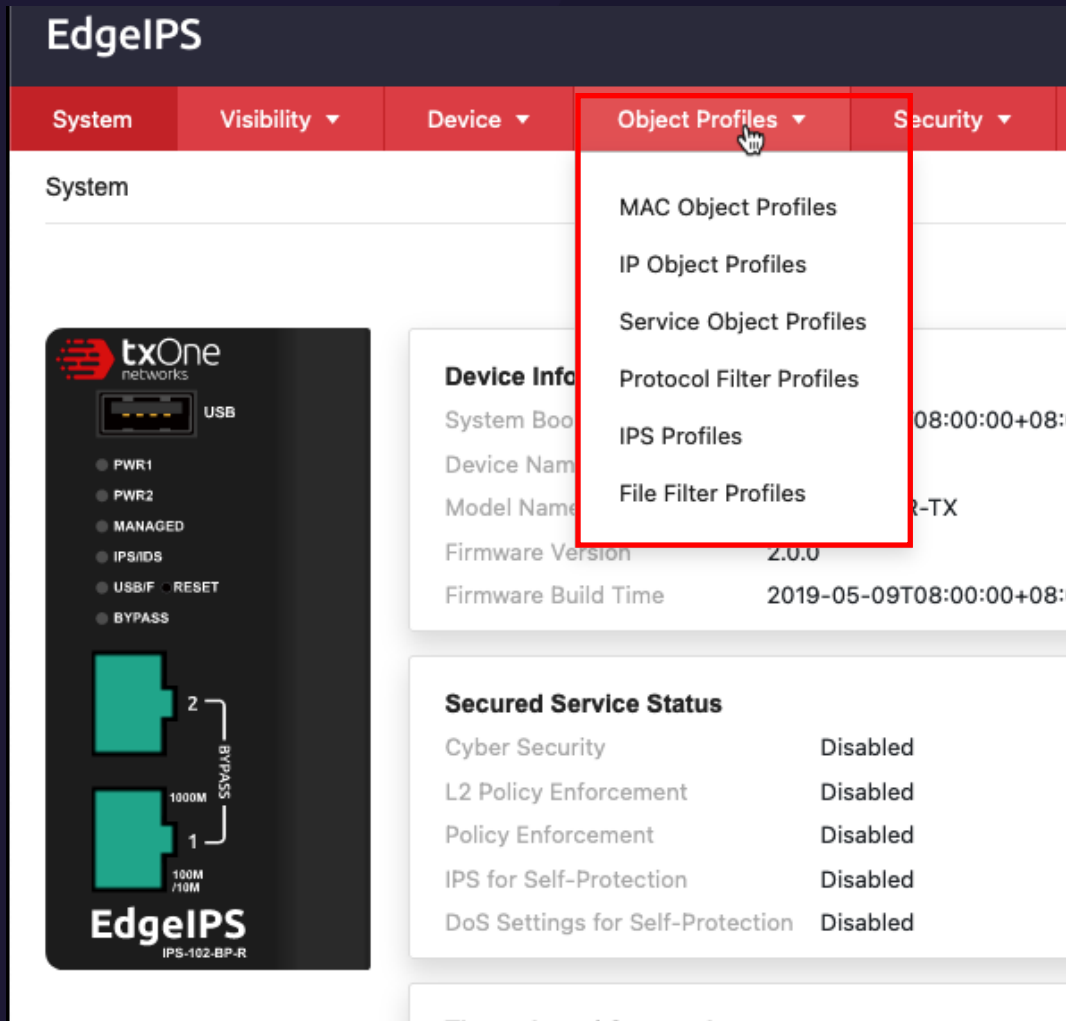
egone.cms.v2_2.6.11.acf

(in the Labkit zip file)

for password use P@ssw0rd
(not 12345678)

Edge Key features and Use-cases

First create: *Object* Profiles Then use a *Policy Enforcement* Profile to enforce these object profiles



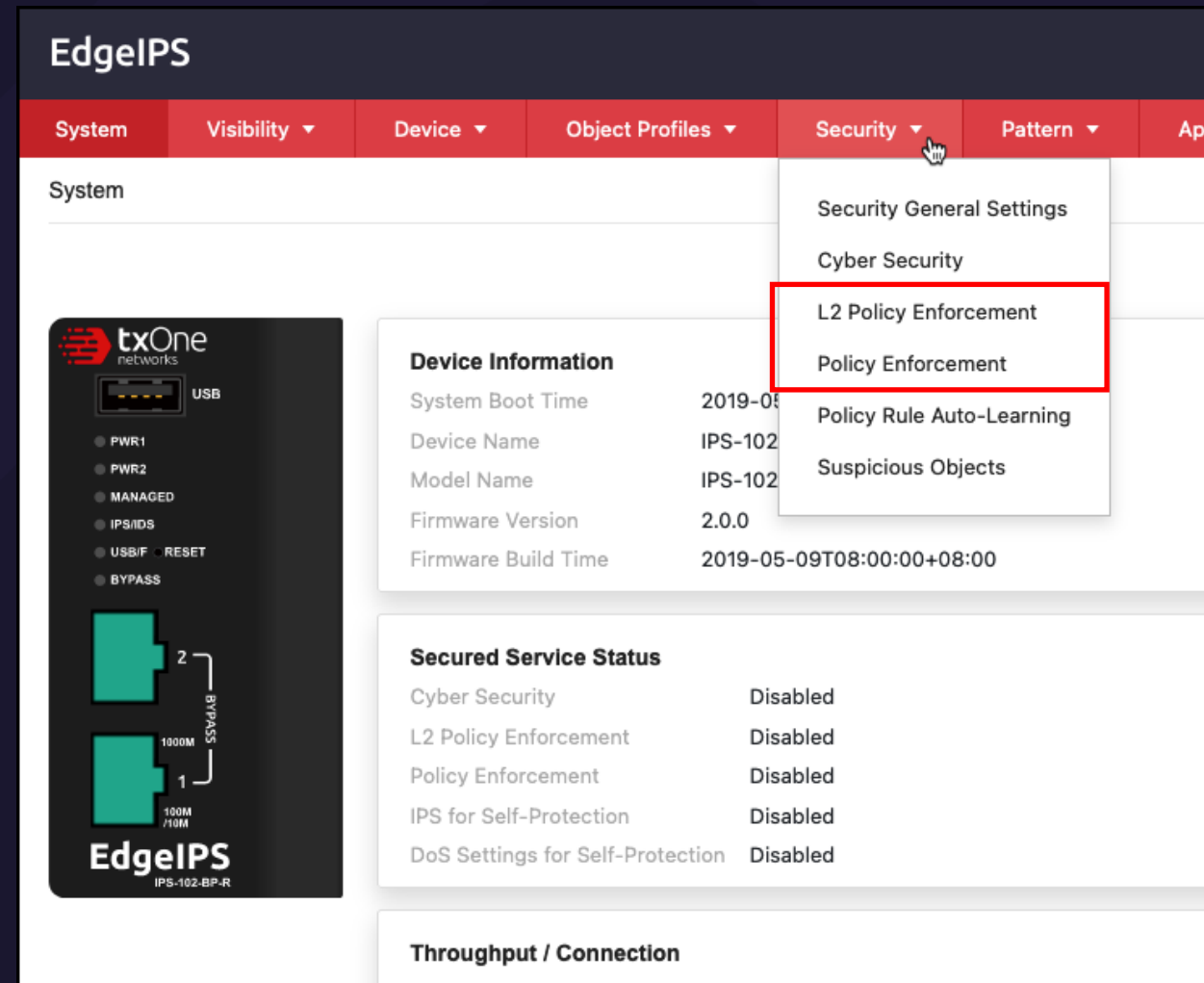
The screenshot shows the EdgeIPS web interface. The top navigation bar includes tabs for System, Visibility, Device, Object Profiles, and Security. The 'Object Profiles' tab is selected, and its dropdown menu is open, showing options: MAC Object Profiles, IP Object Profiles, Service Object Profiles, Protocol Filter Profiles, IPS Profiles, and File Filter Profiles. The 'IPS Profiles' option is highlighted with a red box. The main content area displays 'Device Information' and 'Secured Service Status' for a device named 'IPS-102-BP-R'.

Device Information

System Boot Time	2019-05-09T08:00:00+08:00
Device Name	IPS-102-BP-R
Model Name	IPS-102-BP-R
Firmware Version	2.0.0
Firmware Build Time	2019-05-09T08:00:00+08:00

Secured Service Status

Service	Status
Cyber Security	Disabled
L2 Policy Enforcement	Disabled
Policy Enforcement	Disabled
IPS for Self-Protection	Disabled
DoS Settings for Self-Protection	Disabled



The screenshot shows the EdgeIPS web interface. The top navigation bar includes tabs for System, Visibility, Device, Object Profiles, Security, Pattern, and Application. The 'Security' tab is selected, and its dropdown menu is open, showing options: Security General Settings, Cyber Security, L2 Policy Enforcement, Policy Enforcement, Policy Rule Auto-Learning, and Suspicious Objects. The 'L2 Policy Enforcement' option is highlighted with a red box. The main content area displays 'Device Information' and 'Secured Service Status' for a device named 'IPS-102-BP-R'.

Device Information

System Boot Time	2019-05-09T08:00:00+08:00
Device Name	IPS-102-BP-R
Model Name	IPS-102-BP-R
Firmware Version	2.0.0
Firmware Build Time	2019-05-09T08:00:00+08:00

Secured Service Status

Service	Status
Cyber Security	Disabled
L2 Policy Enforcement	Disabled
Policy Enforcement	Disabled
IPS for Self-Protection	Disabled
DoS Settings for Self-Protection	Disabled

Same settings, but now from EdgeOne

EdgeOne

Dashboard

Visibility ▾

Node Management ▾

Logs ▾

Report

Applications ▾

Administration ▾

About

Node Management > EdgeIPS

Number of Available Node Licenses: 46

Number of Registered EdgeIPS Node Licenses: 2

[Details](#)

Device Groups

Ungrouped (1)

group1 (1)

Devices

Cyber Security

Policy Enforcement

IP Object Profiles

Service Object Profiles

Protocol Filter Profiles

IPS Profiles

File Filter Profiles

Packet Capture

Suspicious Objects

Pattern Settings

+ Add

☐

No.

Profile Name

Se

☐

1

FTP_and_HTTP

TC

EdgeIPS

System

Visibility ▾

Device ▾

Object Profiles ▾

Security ▾

System

txOne networks

USB

PWR1

PWR2

MANAGED

IPS/IDS

USB/F RESET

BYPASS

Device Info

System Boot

Device Name

Model Name

Firmware Version

Firmware Build Time

MAC Object Profiles

IP Object Profiles

Service Object Profiles

Protocol Filter Profiles

IPS Profiles

File Filter Profiles

08:00:00+08:0

R-TX

2019-05-09T08:00:00+08:0

EdgeIPS

System

Visibility ▾

Device ▾

Object Profiles ▾

Security ▾

Pattern ▾

Ap

System

txOne networks

USB

PWR1

PWR2

MANAGED

IPS/IDS

USB/F RESET

BYPASS

Device Information

System Boot Time

Device Name

Model Name

Firmware Version

Firmware Build Time

2019-05-09T08:00:00+08:00

IPS-102

IPS-102

2.0.0

2019-05-09T08:00:00+08:00

Security General Settings

Cyber Security

L2 Policy Enforcement

Policy Enforcement

Policy Rule Auto-Learning

Suspicious Objects

Secured Service Status

Profiles

Profiles

- IP object profiles (IP addresses, IP ranges and subnets)
- Service object profiles (service ports and port-ranges)
- Protocol filter profiles (OT and IT protocols)
- File filter profiles (blocking files containing executable code)
- IPS profiles
- Security profiles (DoS protection)
- MAC address profiles
- Anti-virus profiles
- Domain Name profiles

Service Object Profiles (to be used in *Basic* Filter)

Create Service Object Profile

Profile Name*

FTP_and_HTTP

Description

Service Object List

(Max: 32)

No.1*	TCP	Protocol Number	6	Service Ports	Custom	80	-	80	-
No.2*	TCP	Protocol Number	6	Service Ports	Custom	20	-	21	+ -

Save

Cancel

Service Object Profiles consider the port,
e.g. Port = 80,20,21
regardless of the protocol

Protocol Filter (Object) Profiles (to be used in Advanced Filter)

Edit Protocol Filter Profile

Profile Name*

HTTP

Description

Enabled: 2

Disabled: 64

Total: 66

All Statuses

All Verticals

All Protocols

Q http

☐	Status	Vertical	Protocol	Advanced Setting	Drop Malformed
☐	☒	Common	HTTP	-	

Items: 1-1 / 1

25

per page

1 / 1

<<

<

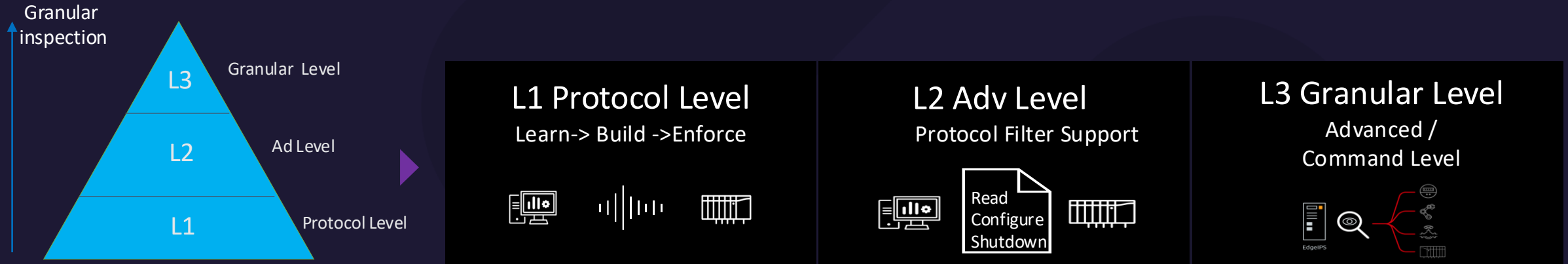
>

>>

Protocol Filter profiles consider the Protocol, e.g. HTTP traffic, regardless of the port

OT Protocol granular inspection solution by vertical

Deep inspection solution - OT protocol Filter



Factory Automation

Modbus(includes UMAS)	GE SDI
CIP(Include PCCC)	GE SRTP
S7Comm	GE CMP
S7Comm +	HartIP
SLMP(MELSEC)	OPC-UA
MELSOFT	MECHATROLINK III
OMRON FINS	EtherCat
TOYOPUC	ROS
Profinet	SimpleMessage
Profinet RT	Emerson ROC
Profinet Safe (I/O)	CODESYS
Profinet DCP	Emerson DeltaV
SECS/GEM	GE-EGD
	GE-iFIX
	TriStation

Power & Electricity

IEC-61850 - GOOSE(*L2)

DNP3

IEC-61850- MMS

IEC-104

ICCP TASE.2

IEC61850-R-GOOSE

IEC60870-R-SV

IEC62056

IEEE C37.118

IEC61850-SV

HealthCare

HL7 / DICOM

Building Automation

BACNet

KNX

CN/IP

Niagara Fox

LonWorks/LonTalk

Common OT Protocol

CoAP

ADDP

RTPS

MDLC

Ether-S-Bus

EtherSIO

Moxa Protocol

OT Protocol granular inspection solution by vertical

0x01: Read Coils
0x02: Read Discrete Inputs
0x03: Read Holding Registers
0x04: Read Input Registers
0x05: Write Single Coil
0x06: Write Single Register
0x07: Read Exception Status
0x08: Diagnostics
0x0B: Get Comm Event Counter
0x0C: Get Comm Event Log
0x0F: Write Multiple Coils
0x10: Write Multiple Registers
0x11: Report Slave ID
0x14: Read File Record
0x15: Write File Record
0x16: Mask Write Register
0x17: Read/Write Multiple Registers
0x18: Read FIFO Queue
0x2B: Encapsulated Interface (MEI) Transport
Custom

L1 Protocol Level

Modbus Advanced Settings

L1 Any
L2 Basic
L3 Advanced Matching Criteria

☒ Read Only ☐ Read / Write ☐ Admin Config

Function List: 0x01: Read Coils

Function Code*: 0x01 ⓘ

Unit ID*: 8 ⓘ

Address*: Single 2 ⓘ

Add

Total Number of Records: 1 (Max: 32)

☐	No.	Function Code	Unit ID	Address
☐	1	(0x01) Read Coils	8	2

OK **Cancel**

HealthCare

ICOM

Automation

onTalk

Common OT Protocol

CoAP
ADDP
RTPS
MDLC
Ether-S-Bus
EtherSIO
Moxa Protocol

File Filter Profile

Edit File Filter Profile

Profile Name*

deny_PE_over_ftp

Description

File Filter Profile Settings

File Filter by Protocol

Protocol*

HTTP

FTP

SMB

Drop the Executable File(s)*

Windows (PE File)

Linux (ELF File)

Zip/gz file(s) larger than 100MB are ignored.

Filter AD GPO

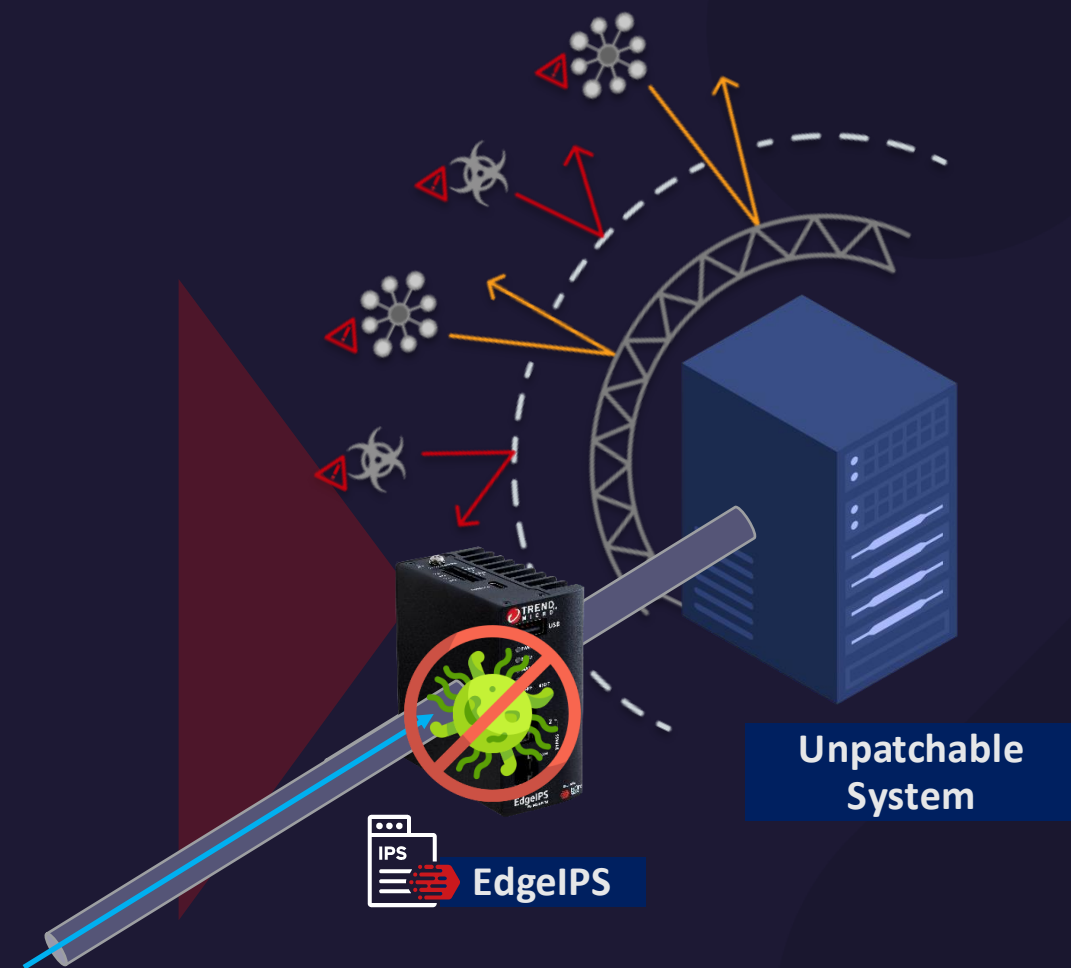
Drop Active Directory(AD) GPO Dispatch

Save

Cancel

IPS profile

Provides "virtual patching" to bridge the ICS Vulnerability Gap



EdgeIPS 102



EdgeIPS Pro 216



EdgeFire



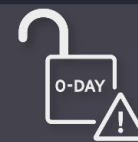
EdgeIPS Pro 1048/2096



Virtual Patching (a "compensating control")



Shield
vulnerable
assets



Prevent
zero-day
attacks



Maintain
Productivity



Enhance the
patching
process

Security Profile (DoS protection)

EdgeIPS

Visibility

Device

Objects & Profiles

Security

Patterns

Applications

Logs

Admin



Security > Security Policies > Cyber Security

☒ Denial of Service (DoS) Prevention  When the Edge device is managed by EdgeOne and the corresponding group settings option in EdgeOne is enabled, the settings here are synced with EdgeOne.

☒ Monitor Mode  ☐ Prevention Mode

☒ TCP SYN Flood	Threshold	10000	Packets 
☒ ICMP Flood	Threshold	10000	Packets 
☒ TCP Port SYN Scan	Threshold	1800	Packets 
☒ TCP Port FIN Scan	Threshold	1800	Packets 
☒ UDP Port Scan	Threshold	250	Packets 
☒ ARP Scan	Threshold	120	Packets 

☒ UDP Flood	Threshold	10000	Packets 
☒ IGMP Flood	Threshold	10000	Packets 
☒ TCP Port NULL Scan	Threshold	1800	Packets 
☒ TCP Port Xmas Scan	Threshold	1800	Packets 
☒ IP Sweep	Threshold	120	Packets 
☒ Ping Sweep	Threshold	120	Packets 

Save

Cancel

Policy Enforcement:

Policy Enforcement Rules

EdgeIPS

Visibility

Device

Objects & Profiles

Security

Patterns

Applications

Logs

Admin

txOne networks

Security > Security Policies > Policy Enforcement

Policy Enforcement General Settings

⚠ When the Edge device is managed by EdgeOne and the corresponding group settings option in EdgeOne is enabled, the sett

Policy Enforcement

Policy Enforcement Operation Mode

Monitor Mode

Default Rule Action

Deny and Log

IPv6 Traffic

Allow

Monitor Mode or Prevention Mode

Policy Enforcement Rule List

Device Rule List

EdgeOne Rule List

+ Create

🗑 Delete

📄 Duplicate

⬆⬆ Change Priority

1 selected

Total Number of Items: 2 (Max: 512)

🔍

1 / 1

<

>

📄

🗑

☐	No.	Status	Rule Name	Connection Direction	Source IP / Object	Source IP / Object Info	Destination IP / Object	Destination IP / Object Info	Service Object	Service Object
☑	1		MyFirstDeviceRule	Bi-directional	User-Defined	10.0.0.15	User-Defined	10.0.0.0/24	Any	Any
☐	2		MySecondDeviceRule	Bi-directional	Any	Any	Any	Any	Any	Any

txOne networks

Policy Enforcement Rules

EdgeIPS

Visibility

Device

Objects & Profiles

Security

Patterns

Applications

Logs

Admin

txOne

networks

Security > Security Policies > Policy Enforcement

Policy Enforcement General Settings

⚠ When the Edge device is managed by EdgeOne and the corresponding group settings option in EdgeOne is enabled, the settings

Policy Enforcement

Policy Enforcement Operation Mode

Monitor Mode

3

Default Rule Action

Deny and Log

IPv6 Traffic

Allow

Policy Enforcement Rule List

Device Rule List

EdgeOne Rule List

1

+ Create

🗑 Delete

📄 Duplicate

↕ Change Priority

1 selected

Total Number of Items: 2 (Max: 512)

🔍

order	No.	Status	Rule Name	Connection Direction	Source IP / Object	Source IP / Object Info	Destination IP / Object	Destination IP / Object Info	Service Object	Service Object
1	1		MyFirstDeviceRule	Bi-directional	User-Defined	10.0.0.15	User-Defined	10.0.0.0/24		
2	2		MySecondDeviceRule	Bi-directional	Any	Any	Any	Any		

This is the **Default** rule.

"Deny and Log" will **only** be applied to traffic that does **not match any of the rules below**

Then, if no match, only then this Default Rule is applied

Default rule:
"Deny and Log" =
- Whitelisting or
- Blacklisting ?

It is Whitelisting
(everything is **denied**,
only traffic matching the
rules is allowed)

TXOne Networks | Keep the Operation Running

Inside a Policy Enforcement Rule

Create Policy Enforcement Rule

Status ☒

Rule Name* ⓘ

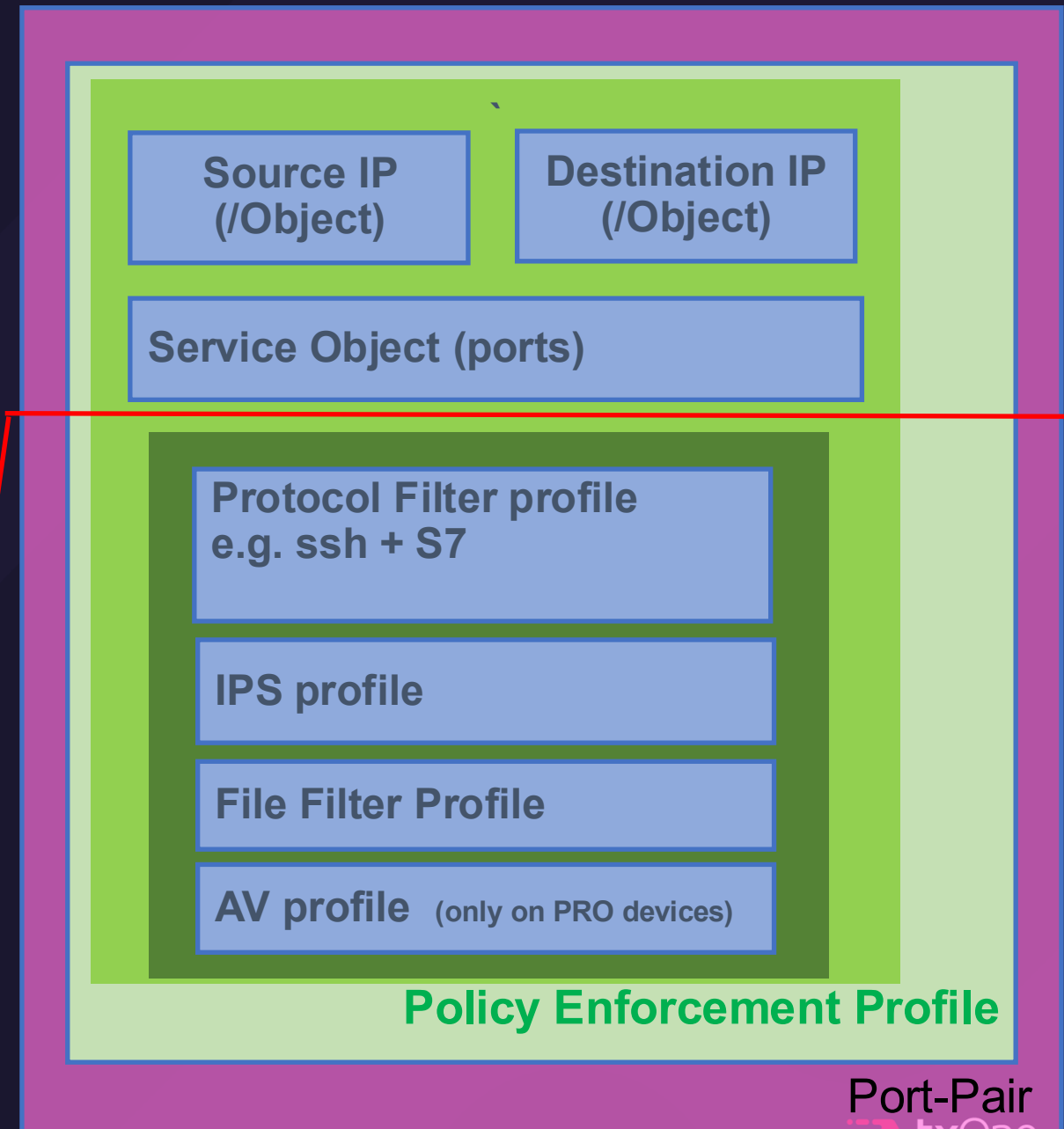
Create Policy Enforcement Rule

Basic Filter

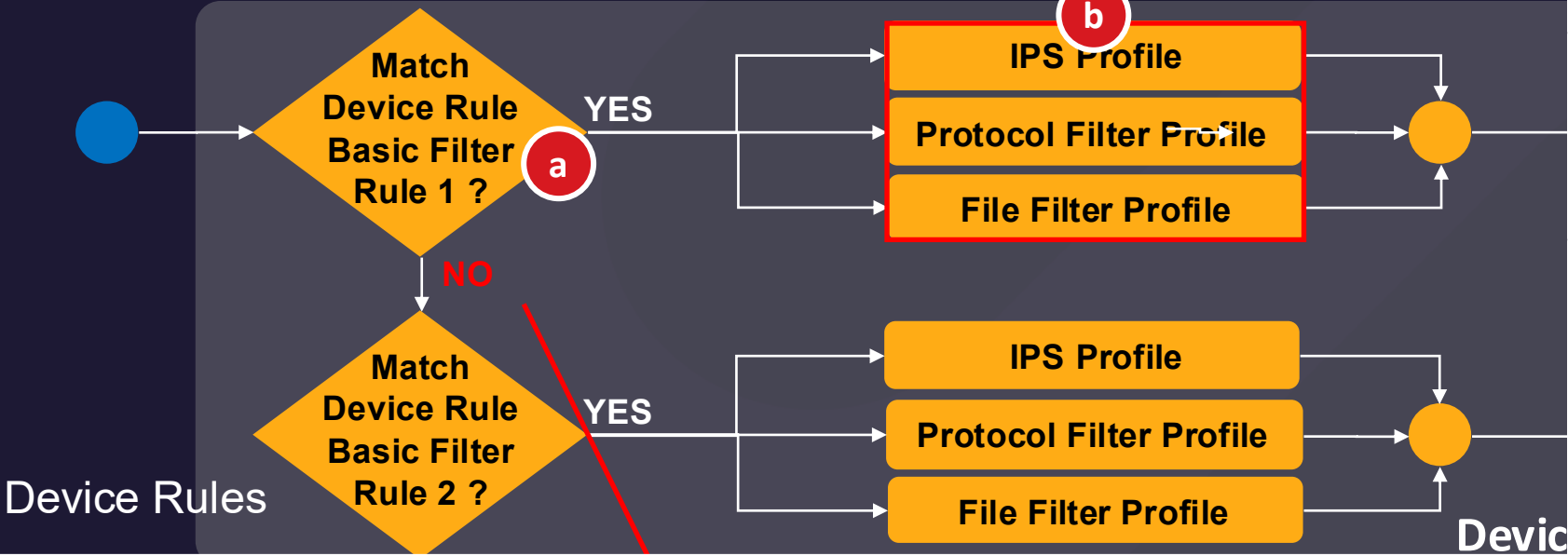
Source IP / Object	IP Object	Engineering
Destination IP / Object	IP Object	BodyShopSubnet
Service Object	Any	
VLAN ID	Any	
Action	Accept	

Advanced Filter ⓘ

☒ Protocol Filter Profile	Factory_Automation
Action	Accept
☒ IPS Profile	Default_IPS_Profile
☐ File Filter Profile	Please select a profile.



Policy Enforcement Rule Decision Flow



EdgeOne Rules

Only the BASIC Filter decides if the rule matches or not

The Advanced Filter is only evaluated if the Basic Filter matches

Inside a Policy Enforcement Rule

Create Policy Enforcement Rule

Status ☒

Rule Name* BodyShopEnforcementForEngine

Create Policy Enforcement Rule

Basic Filter a

Source IP / Object IP Object

Destination IP / Object IP Object

Service Object Any

VI AND ID Any

Action Accept

Advanced Filter b

☒ Protocol Filter Profile Factory_Automation

Action Accept

☒ IPS Profile Default_IPS_Profile

☐ File Filter Profile Please select a profile.

Policy Enforcement Rules:

EdgeIPS

Visibility

Device

Objects & Profiles

Security

Patterns

Applications

Logs

Admin

Security > Security Policies > Policy Enforcement

Policy Enforcement General Settings

⚠ When the Edge device is managed by EdgeOne and the corresponding group settings option in EdgeOne is enabled, the settings here are synced with E

Policy Enforcement

Policy Enforcement Operation Mode

Monitor Mode

Default Rule Action

Deny and Log

IPv6 Traffic

Allow

Policy Enforcement Rule List

Device Rule List

EdgeOne Rule List

+ Create

Total Number of Items: 2 (Max: 512)

1 / 1

No.	Status	Rule Name	Connection Direction	Source IP / Object	Source IP / Object Info	Destination IP / Object	Destination IP / Object Info	Service Object	Service
1		Rule1	Bi-directional	Any	Any	Any	Any	Any	Any
2		Rule2	Bi-directional	User-Defined	10.0.0.15	User-Defined	10.0.0.0/0	Any	Any

Save

Cancel

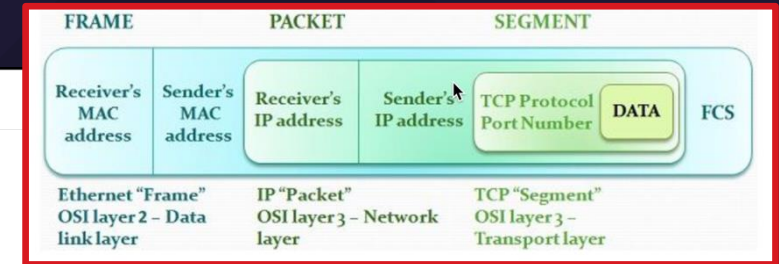
When is the Device Rule, Rule2 evaluated?

Never

We don't even care if rule number 1 is a Deny or an Allow

We know there is a *match* with rule1, so rule number2 will not be evaluated

Processing *inside* a Policy Enforcement Rule:



Create Policy Enforcement Rule

Status ☒

Rule Name*

Description

Basic Filter

Source IP / Object

Destination IP / Object

Service Object

VLAN ID

Action

Advanced Filter

☒ Protocol Filter Profile

Action

☐ IPS Profile

☐ File Filter Profile

OK Cancel

Demo **Policy Rule Auto-Learning**

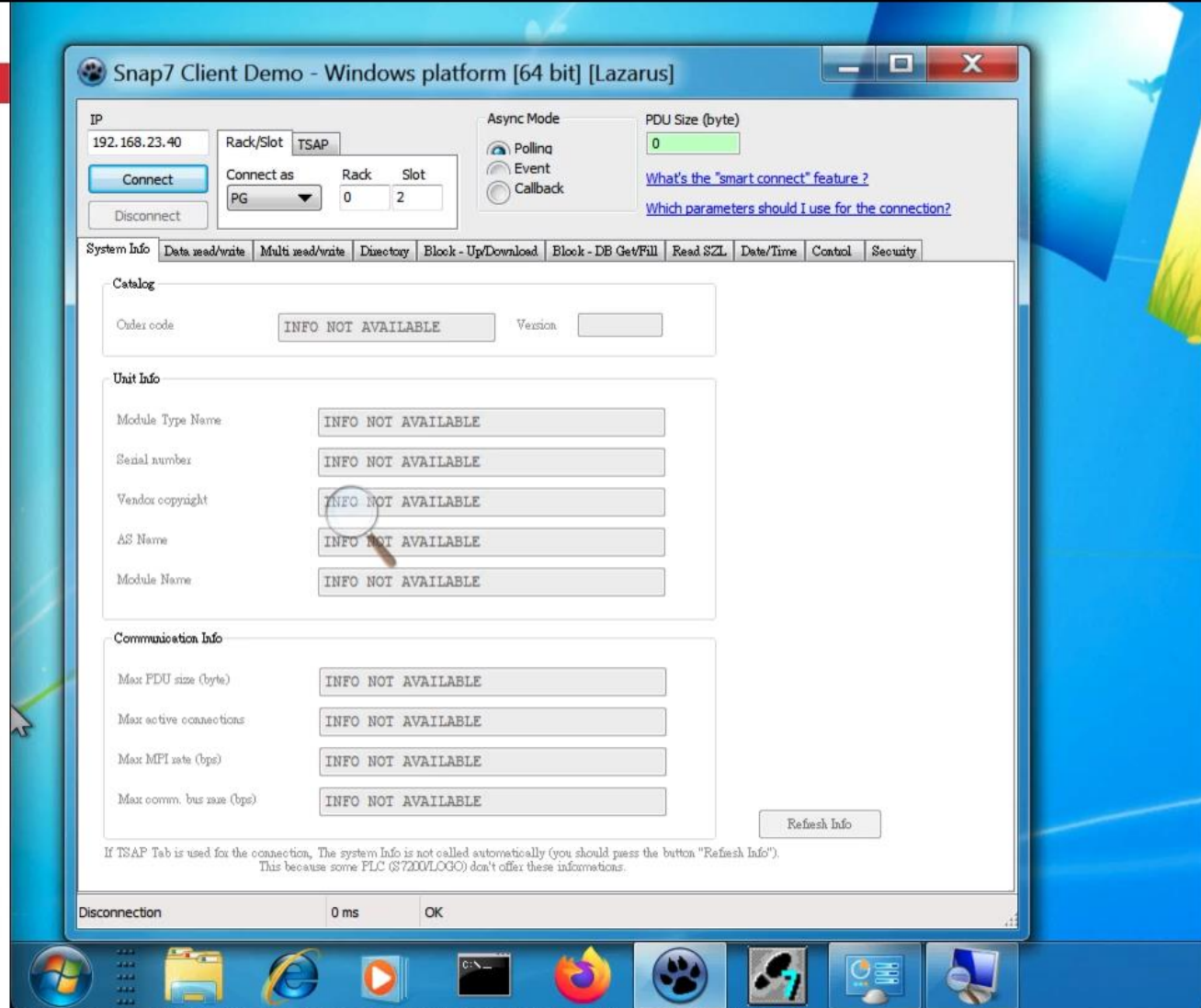
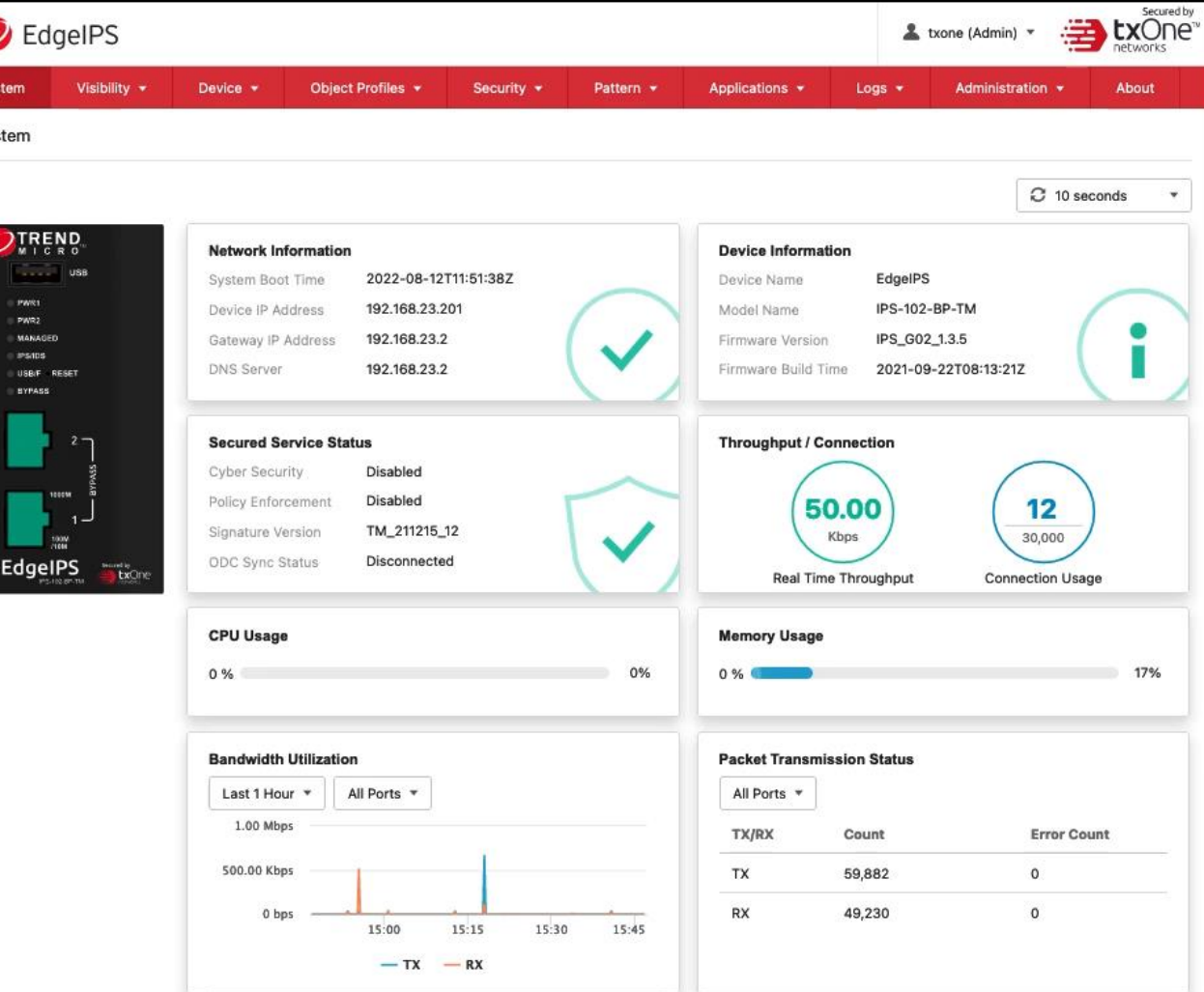
(recap)

Before Policy Rule Auto-Learning: any device can talk to any device

After Policy Rule Auto-Learning: only whitelisted communications are allowed"

-> Micro-segmentation has been enforced

Demo: Policy Rule Auto-Learning



```
2022-08-12 17:46:37 [192.168.23.26] System clock read requested
2022-08-12 17:46:38 [192.168.23.26] System clock read requested
2022-08-12 17:46:38 [192.168.23.26] System clock read requested
2022-08-12 17:46:39 [192.168.23.26] System clock read requested
2022-08-12 17:46:42 [192.168.23.26] Client disconnected by peer
```

Copy Policy Enforcement Rule Set

The screenshot displays the txOne EdgeOne management interface. The top navigation bar includes 'EdgeOne', 'Visibility', 'Nodes', 'Logs & Reports', 'Applications', and 'Admin'. The breadcrumb path is 'Nodes > EdgeFire Family > EdgeFire Std'. A 'Global Policy Rule Settings' button is visible in the top left. The left sidebar shows a list of 'Edge Groups' (Ungrouped, Group Name A through O) and a 'Group Settings' panel for 'Group Name A'. The main area shows 'All Edge Devices (6)' with a grid of device icons. A context menu is open over one device, listing actions such as 'Access Remote Edge Device', 'Copy IP Address', 'Copy MAC Address', 'Edit', 'Configure Port Security', 'Upgrade Firmware', 'Reboot', 'Switch Firmware Versions', 'Delete', 'Change Group', 'Copy Policy Enforcement Rule Set' (highlighted with a red box), and 'Collect Logs for Debugging'. A detailed view of device 'IEF1014-R' is shown on the right, displaying fields like Serial Number, Hostname, Location, IP Address, MAC Address, and Model.

Are my rules being hit ?

EdgeOne

Visibility

Nodes

Logs & Reports

Applications

Admin

Nodes > EdgelPS Family > EdgelPS Std

Global Policy Rule Settings

Group Name Search

Edge Groups

Group Settings

Ungrouped (6)

Group_A (6)

Group_B (6)

Group_C (6)

Group_D (6)

Group_E (6)

Group_F (6)

Group_G (6)

Group_H (6)

Group_I (6)

Device List

Edge Device General Settings

Edge Device Security Settings

Policy Rule Auto-Learning

Cyber Security

EdgeOne Security Settings

CPSDR Rules

CPSDR Threshold Settings

Policy Enforcement

Domain Name Filter Rules

EdgeOne Objects & Profiles

IP Objects

Service Objects

Protocol Filter Profiles

IPS Profiles

File Filter Profiles

The group applies the global policy rules. If you want to configure the settings, please go to the [Global Policy Rules](#) to make the changes.

Policy Enforcement General Settings

Policy Enforcement

Policy Enforcement Operation Mode Monitor Mode

Policy Enforcement Default Rule Action Deny and Log

IPv6 Traffic Accept

Policy Enforcement Rule List

Click to [learn more](#) about how different policy rules are applied.

Create

Group: 1 | Global: 3 | Total: 4 (Max:512) 1 / 1

All Hit Count Search

No.	Status	Rule Name	Hit Count	Source IP / Object	Source IP / Object Info
1	☒	EdgeOne_PE_RuleName_1	123	Any	Any
2	☒	Global_RuleName_1	0	Any	Any
3	☒	Global_RuleName_2	91	Any	Any
4	☒	Global_RuleName_3	9990+	Any	Any

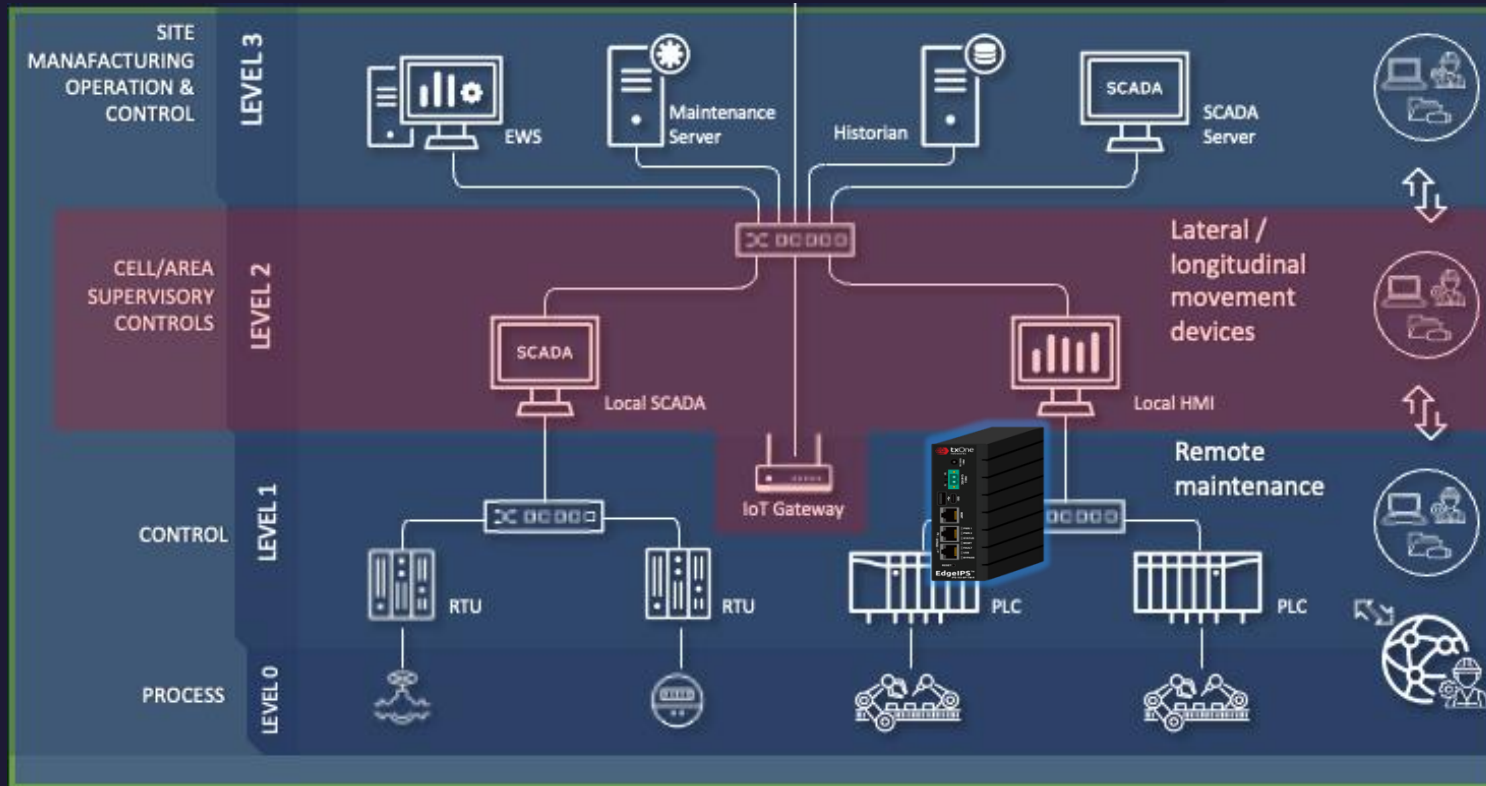
Hit Count Details (Global_RuleName_2)

Total Number of Items: 3 1 / 1

Edge Hostname	IP Address	MAC Address	Related Asset Count	Hit Count
EdgelPS-103	10.24.40.141	00:90:0b:df:8f:21	30	70
EdgelPS-103F	10.24.40.140	00:90:0b:df:8e:ca	10	20
EdgelPS-102	10.24.40.138	08:35:71:2f:59:86	1	1

Close

How long or how much traffic does it take to get a good "learned" profile? (recap)



Scenario A

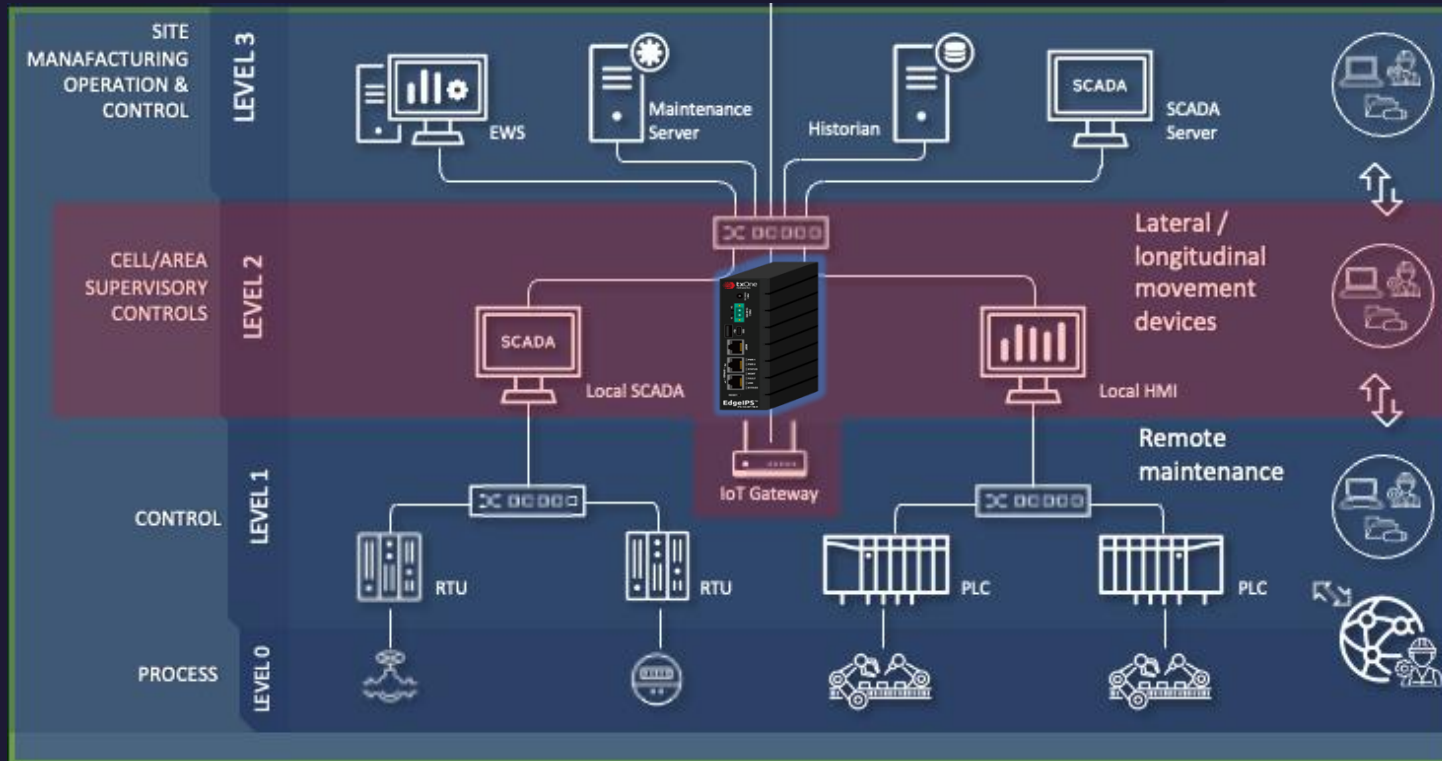
Scope:

- There is only device **PLC** behind EdgeIPS

Operation Cycle

- PLC: Repeat operation once a hour
- Auto Rule Learning period
- Apply at least one hour for a good learned results.

How long or how much traffic does it take to get a good "learned" profile? (recap)



Scenario B

Scope

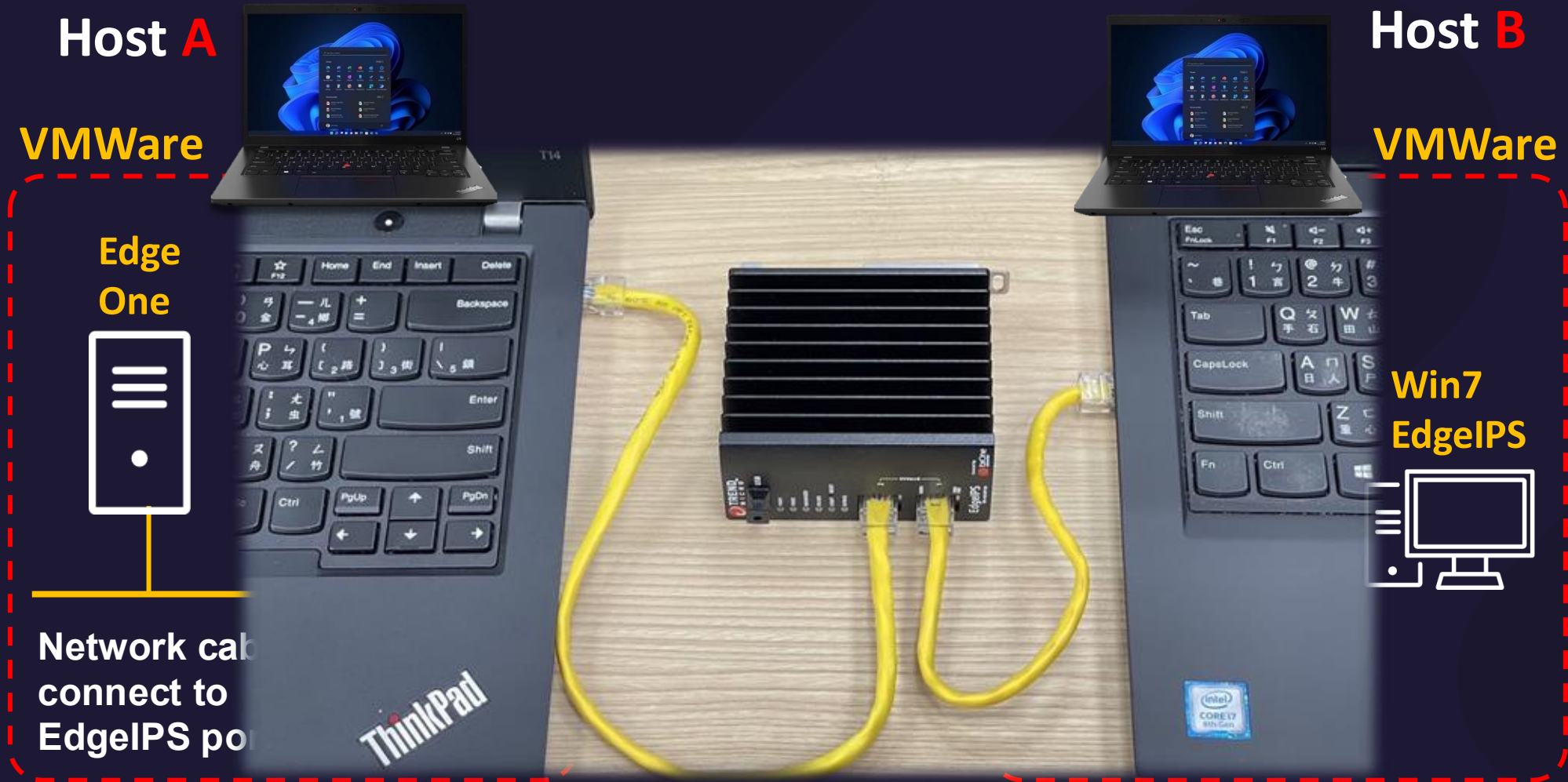
- There are 2 devices connected to the IoT Gateway, and we install EdgeIPS in front of a IoT gateway.

Operation Cycle

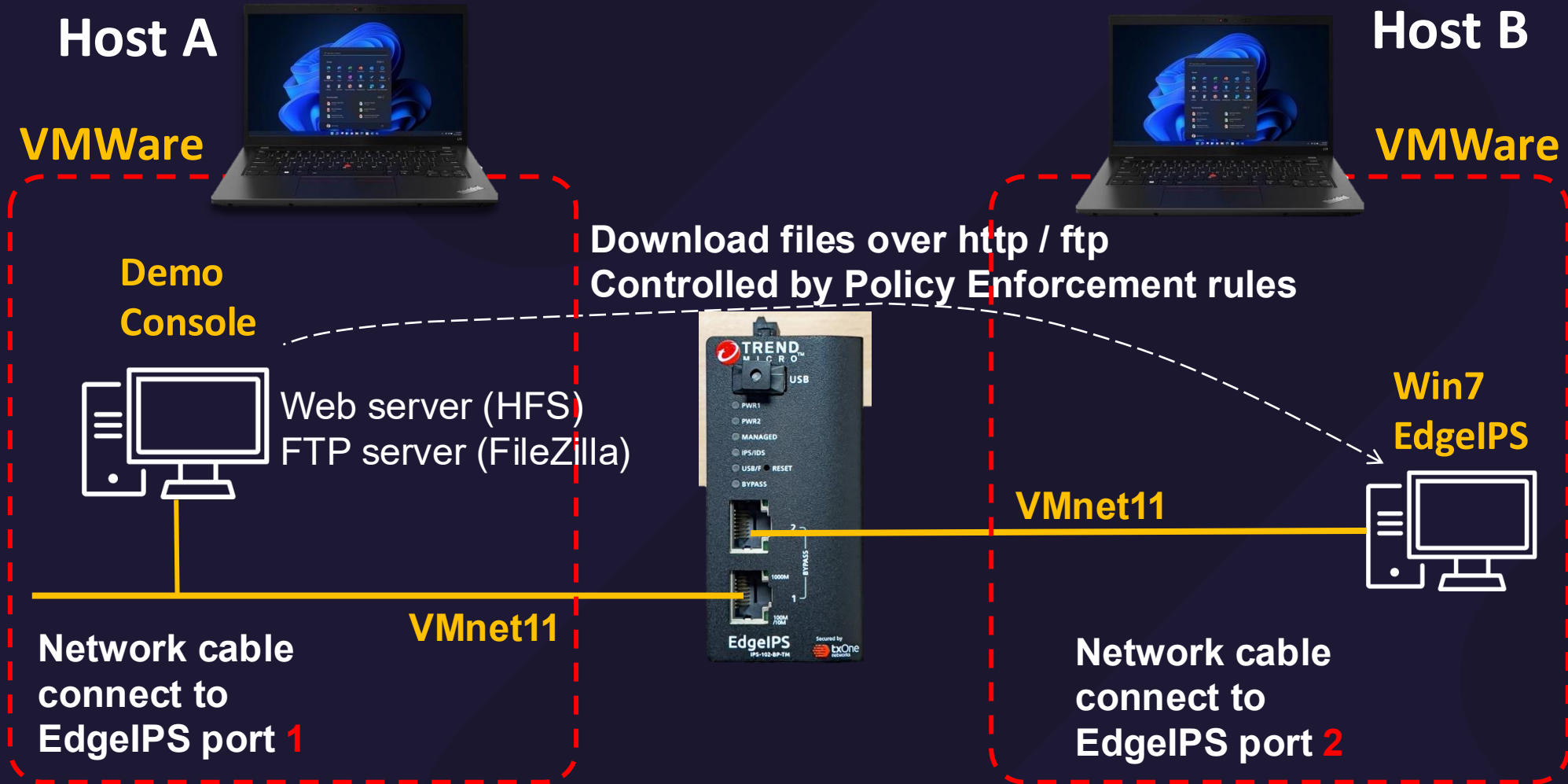
- Device A: Repeat operation once a day
- Device B: Repeat operation cycle once a week
- Auto Rule Learning period
 - Apply at least **one week** for a good learned results.

Edge configuration hands-on labs

Lab lay-out for lab 7 and later



301 – Lab7 Download Files over HTTP and FTP



301 - Lab7 Structure

Download file over FTP

Download file over http

Log Purge

Demo
Console VM



Web server + FTP server



Win7
EdgeIPS VM

Download files

Repeat the download of files but with different policy configurations on EdgeIPS

PE-in-zip.zip : Zip file which contains PE files

Text-file.txt : Clear text file

Lab sub tasks

Lab main tasks

Monitor mode vs.
Prevention Mode

Download is allowed
Download is blocked

Basic Filter

Allow only designated IP addresses and ports

IPS Profile

PE file downloads over http / ftp are identified by EdgeIPS as **Suspicious PE activities**

Protocol Filter Profile

Only **Allow** ftp protocol

File Filter Profile

Deny PE file transmission over ftp

A note on Policy Rule Auto Learning

- If a packet **matches** an existing Policy Enforcement rule, it **will not** be analyzed by the auto learning module.
- (this allows us to run multiple learning sessions over time. Only "new traffic will be learned")
- Keep in mind that a "match" can be a Deny or an Allow

Labs Edge Key features

- [301 EdgeIPS LabGuide v.....pdf](#)
 - Lab7: Policy Enforcement Configuration
 - Lab8: edgeIPS Policy Rule
- [401 EdgeIPS LabGuide v..... pdf](#)

Start at section 3.2 EdgeIPS Environment Preparation (create IPS rule !)

- Lab1: Protection Against **WannaCry Ransomware** Attack.
- Lab2: Asset Vulnerability Management .
- Lab3: EdgeIPS Virtual patching to protect legacy assets
- Lab4: EdgeIPS Policy Rule Auto-Learning
- Lab5: EdgeIPS OT Workflow Whitelisting
- Lab6: EdgeIPS Domain Name Filter
- Lab7: CPSDR Networking
- ~~• Lab8: EdgeFire Client-to-Site VPN Configuration~~
- ~~• Lab9: EdgeFire Site-to-Site VPN Configuration~~
- ~~• Lab10: EdgeFire High availability Configuration~~

Suspicious Objects & integrations

Suspicious Objects

3rd Party Solutions
(Suspicious Object Source)



API call

ODC-VA



FW 1.3 or later
version

SO list Table for
IPS
Configured by ODC



SO Type: Node, Link

EdgeIPS

EdgeIPS



FW 1.2 or later
version

SO list - Table
in SO Feature Page



SO list

OT Defense Console

txadmin (Admin)

Secured by txOne networks

Dashboard Visibility Node Management Logs Report Applications Administration About

Applications > Suspicious Object Pool

Suspicious Object List Settings

Packet Capture
Suspicious Object Pool

Total Number of Pools: 3

All Statuses All Types All Sources All Risk Levels Search

Status	ID	Type	Source	Object Content	Risk Level	Last Update
New	94142a67b90cfc6b69d4a525e14fc5a62	Link	chrisV	Src IP=61.100.32.90, Dst IP=61.100.32.92, Dst Port=8080	Medium	2022-06-25T
New	ed065abc16afa02c3209074043965924	Node	chrisV	MAC=AA:BB:CC:DD:EE:FF	Medium	2022-06-25T
New	fdf3c5a27dc580e224a400fd732766be	Node	chrisV	IP=61.100.32.91	Medium	2022-06-25T

EdgeIPS

txone (Admin)

Secured by txOne networks

System Visibility Device Object Profiles Security Pattern Applications Logs Administration About

Security > Suspicious Objects

Suspicious Object General Setting

☒ Suspicious Object ☐ Enabling this feature and Suspicious Object priority is higher than Policy Enforcement.

Suspicious Object Operation Mode ☒ Monitor Mode ☐ Prevention Mode

Suspicious Object Rule List

Drop: 3 Bypass: 0 Total: 3

All Types All Sources All Risk Levels All Actions Search

No.	ID	Type	Source	Object Content
1	94142a67b90cfc6b69d4a525e14fc5a62	Link	chrisV	Src IP=61.100.32.90, Src Port=Any, Dst IP=61.100.32.92, Dst Port=8080, Proto
2	ed065abc16afa02c3209074043965924	Node	chrisV	AA:BB:CC:DD:EE:FF
3	fdf3c5a27dc580e224a400fd732766be	Node	chrisV	61.100.32.91

Third-party Integrations

SIEM platform partners



ODC admin configures the syslog settings:

- server address & port
- protocols: TCP/UDP
- format: CEF/LEEF
- facility level
- log level
- log types

TXOne ODC



ICS detection partners



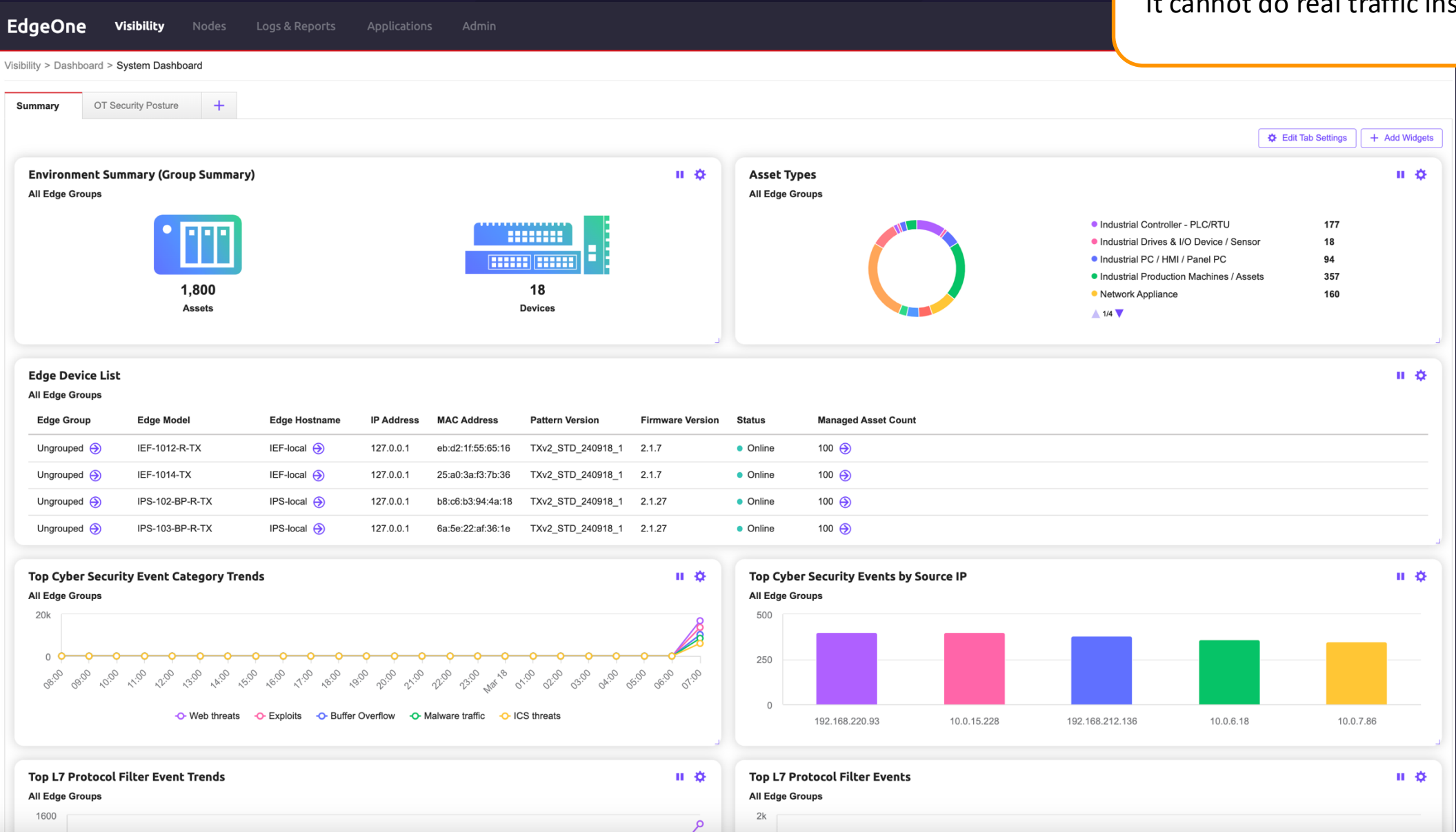
- 1 User configures the ODC IP address in the 3rd-Party products
- 2 The 3rd-Party products notify the ODC to block nodes or links via RESTful APIs
- 3 Once confirmed, the block-lists will be added to the Edge devices



EdgeOne Simulator

EdgeOne Simulator / Demo

edgeOneDemo2.4.15.zip
This is an UI ***Simulator*** with fake events.
It cannot do real traffic inspection





txOne™
networks

